



Society of Petroleum Engineers

SPE-227332-MS

Stimulation Workflow for an HP/HT High-Anisotropy Carbonate Reservoir in Iraq: Coiled Tubing Wellbore Cleaning with Customized Solvent Blend to Fine-Tune Stimulation Design Using Perforation Sealer Diversion Technique

A. Masoodi, Halliburton, Basra, Iraq; J. R. Tellez, Halliburton, Dhahran, Saudi Arabia; M. Ramirez Nuñez and J. Corrales, Halliburton, Basra, Iraq; X. A. Alarcon, Halliburton, Dubai, United Arab Emirates; A. Ghonaim, Halliburton, Basra, Iraq; M. De Leonardis, Halliburton, Dubai, United Arab Emirates; B. Pacha, S. Stolyarov, A. Kadhim, and A. Ihsan, Kuwait Energy Basra Limited, Basra, Iraq

Copyright 2025, Society of Petroleum Engineers DOI [10.2118/227332-MS](https://doi.org/10.2118/227332-MS)

This paper was prepared for presentation at the Middle East Oil, Gas and Geosciences Show (MEOS GEO), Manama, Bahrain, 16 – 18 September 2025.

This paper was selected for presentation by an SPE program committee following review of information contained in an abstract submitted by the author(s). Contents of the paper have not been reviewed by the Society of Petroleum Engineers and are subject to correction by the author(s). The material does not necessarily reflect any position of the Society of Petroleum Engineers, its officers, or members. Electronic reproduction, distribution, or storage of any part of this paper without the written consent of the Society of Petroleum Engineers is prohibited. Permission to reproduce in print is restricted to an abstract of not more than 300 words; illustrations may not be copied. The abstract must contain conspicuous acknowledgment of SPE copyright.

Abstract

The primary objective of this case was to optimize well performance and enhance hydrocarbon recovery from a complex, high- pressure/high-temperature (HP/HT), low-permeability carbonate reservoir. This approach also addressed asphaltene solidification in the wellbore that blocked access to the perforations. Success resulted from the combined application of coiled tubing (CT), a tailored emulsified solvent system (ESS), and subsequent bullhead stimulation. Conventional acid and appropriately sized perforation sealers improved stimulation effectiveness to divert the treatment to low-permeability layers.

In response to the complex reservoir characteristics and HP/HT environment, the ESS with a high-dispersant load strategy was deployed for wellbore cleanup using CT and a high differential jetting nozzle. Acid stimulation and diversion design were carefully optimized to ensure maximum coverage in the 52-m interval affected by high anisotropy under extreme HP/HT conditions. Extensive simulations guided the development of a customized diversion solution to enhance stimulation in low-permeability zones. The combined effect of CT and the high-dispersant solvent system enabled thorough wellbore cleanup, effective treatment, and increased hydrocarbon recovery.

The integrated workflow led to substantial improvements in hydrocarbon production. The high-dispersant ESS effectively dissolved solidified asphaltenes and restored full access to the perforated interval. Perforation sealers enabled precise diversion and optimized acid penetration into the low-permeability zones. As a result, production rates rose from a completely blocked reservoir to 3,500 BOPD, which confirmed the effectiveness of this synergistic approach. This case study highlights the robustness of the solvent blend, accuracy of the jetting nozzle deployment, and versatility of the diversion technique and establishes a new benchmark for HP/HT carbonate reservoir stimulation.

The innovative integration of high-dispersant solvent systems, advanced jetting nozzle deployment, and precise diversion techniques are highlighted in this paper. A repeatable methodology is presented to improve well performance under extreme HP/HT conditions by effectively addressing asphaltene challenges using

stimulation optimization in complex, anisotropic reservoirs. The discoveries provide valuable insights into asphaltene remediation, deliver actionable advancements in reservoir management practices, and contribute to more efficient and sustainable hydrocarbon recovery in difficult environments.

Introduction

Carbonate reservoirs in Iraq display extreme heterogeneity and anisotropy. Some zones are vuggy, others include natural fractures, and still others remain impermeable. This geological complexity creates dual-porosity behavior and unpredictable fluid pathways. Consequently, stimulation fluid preferentially flows into high-permeability channels and leaves tight sections of the reservoir untreated. Key reservoir measurements, such as porosity, permeability, and pressure, reflect only the most permeable streaks and yield inaccurate baselines for production forecasts.

The stimulation of HP/HT carbonate formations compound these challenges. Elevated temperature accelerates acid-rock reactions, which forces operators to choose either extremely high pump rates (often beyond CT limits) or specialized retarded-acid systems to slow reaction kinetics and achieve deeper penetration. Under HP/HT conditions, crude oil's asphaltene fraction loses stability. Asphaltenes are large, nonvolatile polar molecules that remain dissolved under reservoir pressure and temperature. A reservoir pressure drop below the asphaltene onset pressure (AOP) or a temperature decline reduces solubility. As a result, asphaltenes flocculate into nanoscale chain structures and eventually deposit as solid aggregates.

Scale formation further complicates HP/HT operations. Sharp drops in pressure and temperature, along with high solid content from drilling and completion fluids, promote scale deposition in perforations and the near-wellbore (NWB) region. This solid buildup reduces effective permeability, distorts pressure readings, and skews production estimates.

HP/HT carbonate reservoirs provide strong economic potential despite these significant limitations. The study well exhibits multiple, interrelated challenges that include:

- Scale layers near the perforations
- Current NWB formation damage
- Long pay intervals with dramatic permeability contrasts
- Asphaltene buildup in the completion string, which impedes slickline, wireline, and CT access
- HP/HT conditions that render conventional 15% HCl acid impractical at CT flow rates

Successful operations require an unobstructed flow path from the surface to the formation. Essential actions include:

- Effective removal of scale and asphaltene deposits through targeted solvents or tailored acid/solvent formulations
- Implementation of a strong diversion strategy that redirects fluid into low-permeability zones and minimizes facial dissolution—"In HP/HT carbonates, high temperatures cause HCl to react rapidly at the rock face, limiting wormhole propagation to just 1 ft without proper diversion"
- Validation of CT pump rates that balance wormhole depth with operational safety under HP/HT conditions

A comprehensive stimulation plan requires full evaluation of the following factors:

- Asphaltene destabilization, deposition mechanisms, and mitigation techniques
- HP/HT and H₂S interactions and their impact on fluid chemistry and materials selection

- CT intervention limitations in highly anisotropic formations and strategies for optimized fluid placement

Asphaltene Deposition

Asphaltenes are large, natural, nonvolatile polar molecules within crude oil under HP/HT reservoir conditions. Their solubility decreases whenever temperature or pressure drops, which causes them to flocculate into nanoscale chain structures that eventually form solid deposits.

Asphaltene precipitation causes severe damage in the NWB region, where it plugs pore throats and reduces permeability in proportion to deposition volume. In tight carbonate zones, even minimal deposition can stop flow completely. Effective mitigation strategies require the evaluation of the AOP before intervention and, if necessary, coinjection of solvents or dispersants with acid to maintain asphaltenes in the solution (Fig. 1 and Fig. 2).

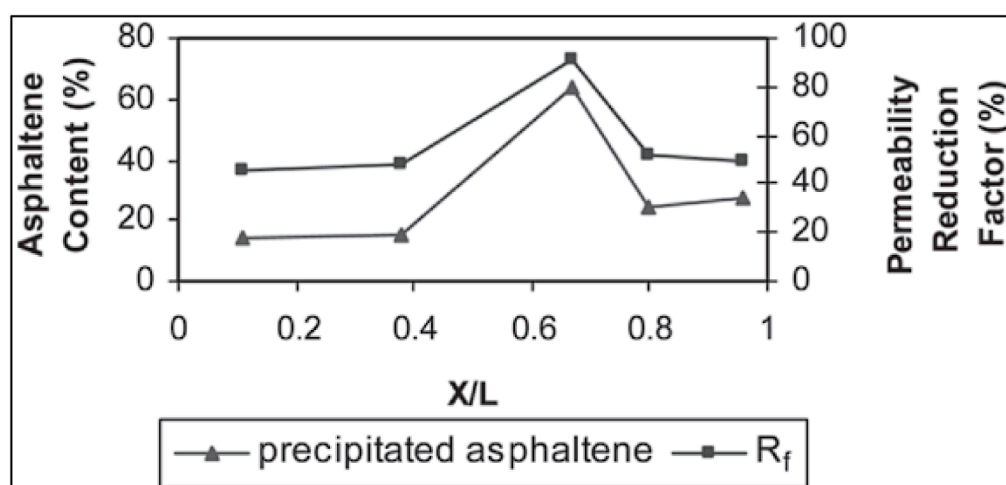


Figure 1—Relationship between precipitated asphaltene and permeability-reduction factor indicates that increased precipitation correlates with greater permeability loss.

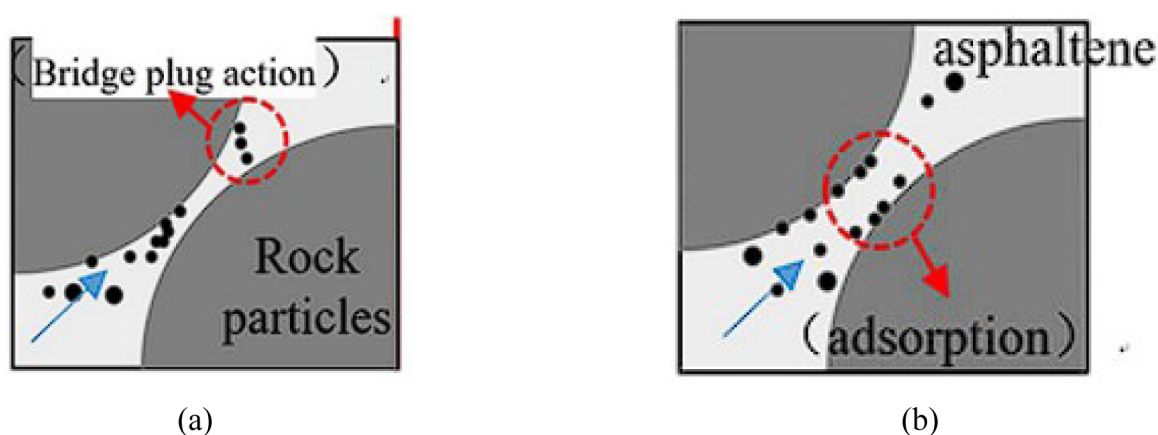


Figure 2—Pore throat plug caused by asphaltene precipitation: (a) low-permeability zone plug because of asphaltene; (b) particle adsorption by the rock in a medium-permeability zone and eventual plug.

Two key measurement points define asphaltene behavior, detailed in the next two subsections (AOP and ADP).

Asphaltene Onset Pressure (AOP)

AOP defines the pressure (at a fixed temperature) where dissolved asphaltene molecules in crude oil begin to form nanoscale aggregates (flocs). Reservoir conditions greater than AOP provide sufficient solvency from dissolved gas and thermal energy to keep asphaltenes stably dissolved. As pressure declines, particularly near the bubblepoint, gas exsolves, which reduces asphaltene solubility and triggers aggregation. Once pressure crosses AOP, loosely bound nano aggregates form. Further pressure decline causes floc growth and potential deposition on rock surfaces, tubular walls, or flowlines.

Thermodynamic Basis. Under high pressure, the oil's solvent power is bolstered by gas in solution. When pressure falls lower than AOP, the solvent power weakens faster than the tendency for asphaltene molecules to associate (molecular chain-molecular chain) through stacking and polar interactions. This imbalance causes flocculation. A phase diagram often illustrates this behavior, where AOP marks the boundary between a stable single-phase region and a two-phase region where asphaltene precipitation occurs.

Measurement Techniques. Several laboratory methods provide reliable means to determine AOP and understand precipitation behavior under reservoir conditions.

- Optical Cell/AFP Tests: Fluid samples are placed in a high-pressure view cell. Pressure is gradually reduced at a constant temperature while turbidity or light scatter is monitored. The pressure where turbidity first appears is recorded as AOP.
- Reaction-Limited Aggregation (RLA) and Diffusion-Limited Aggregation (DLA): These kinetic models help predict the onset of flocculation by modeling how quickly asphaltene molecules collide (DLA) or react (RLA) under pressure and temperature changes. Simulations using these models can estimate AOP without extensive experiments.
- Titration Methods: Precipitation agents (e.g., n-alkanes) are incrementally added to crude samples. The concentration at which visible flocculation occurs correlates with equivalent pressure conditions.

Field Implications. A pressure drop lower than AOP produces several adverse effects:

- Formation Damage: Nano aggregates block pore throats and reduce permeability, particularly critical in low-permeability carbonate zones.
- NWB and Surface Deposition: Aggregated asphaltenes may deposit an "organic skin" on casing, tubing, and flowlines that raises frictional pressure, hinders flow, and intensifies pressure reduction, which promotes further asphaltene deposition.
- Well Intervention Risks: Flocculated particles can adhere to CT, slickline, or wireline tools and increase the likelihood of mechanical sticking during remediation.

Asphaltene Disappearance Pressure (ADP)

ADP defines the pressure, at a fixed temperature, where previously precipitated asphaltene solids redissolve back into the oil phase when pressure increases. In essence, ADP marks the transition from a two-phase region (oil and asphaltene solids) to a stable single-phase region. As pressure rises above ADP, solvent power is restored. Dissolved gas reenters the solution, which increases asphaltene solubility and causes solid aggregates to shrink and vanish.

Thermodynamic Context. ADP typically lies at or slightly above the saturation (bubblepoint) pressure. Crude composition and temperature determine the exact position. When pressure is restored above ADP, exsolved gas redissolves, which increases oil density and solvency. This resolution process reverses flocculation and clears asphaltene deposits previously formed between AOP and ADP.

Measurement and Hysteresis. In practice, during a controlled pressure cycle (decline then rise), the pressure at which asphaltene solids visually disappear is recorded as ADP, often observed in the same high-pressure optical cells used for AOP tests. Because of hysteresis, ADP is usually lower than the original AOP, since redissolution requires less aggressive conditions than initial precipitation.

Operational Significance. A logical understanding of AOP and ADP provides critical guidance for production management and for the selection of appropriate remediation strategies in HP/HT reservoirs.

- **Dynamic Reservoir Management:** Accurate knowledge of AOP and ADP allows operators to design pressure management schemes (e.g., controlled drawdown, gas reinjection) that help prevent extended periods less than AOP or to reactivate wells through a pressure increase greater than ADP.
- **Mitigation Strategies:** If reservoir pressure falls less than ADP, asphaltene solids can redissolve, which temporarily alleviates skin damage. Operators often cycle pressure in a controlled manner (e.g., gas injection to raise bottomhole pressure) to reach ADP and remove deposits.
- **Design of Inhibitors and Dispersants:** Knowledge of AOP/ADP windows guides the selection and concentration of chemical inhibitors or dispersants so that asphaltenes remain stable even when pressure fluctuates.

An overview of the various stages of asphaltene assembly structures has been described. Yen (1994) categorized the asphaltene assembly into six structural orders: unit sheets, stack association, aggregation, assemblages, clusters, and flocs/spherules. Each structural order was expressed by an approximate size. Unit sheets, the smallest species, measure between 1.2 to 2.0 nm, aggregates measure approximately 5 nm, and clusters range from 200 to 2,000 nm. Based on multiple independent investigations that describe asphaltene association, an overall equilibrium is shown in Fig. 3.

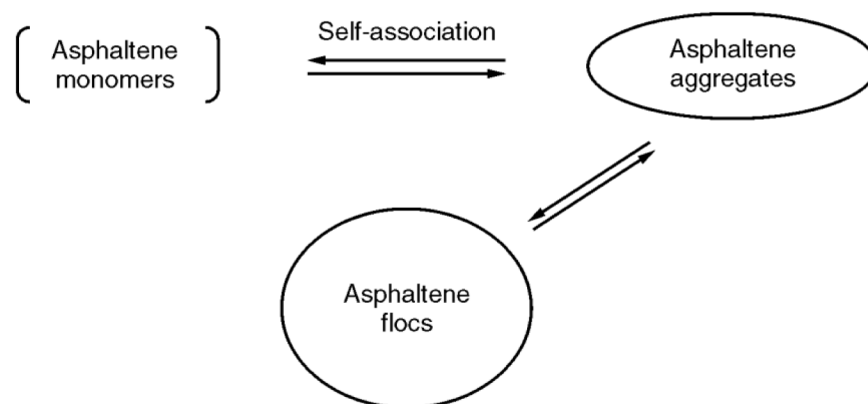


Figure 3—Asphaltene life cycle from monomers to flocs.

Several methods control asphaltene, such as a reservoir pressure increase through CO injection and asphaltene inhibitor injection into the wells. However, those approaches are not the focus of this paper; this discussion centers on fluid customization to dissolve current asphaltene and to delay precipitation to refine the acid stimulation design.

HP/HT and High H₂S Challenges

As highlighted previously, temperature and pressure significantly influence asphaltene deposition. Additional challenges under HP/HT and sour conditions include the following.

Corrosion and Tubular Integrity

Elevated temperature and pressure under HP/HT conditions accelerate the corrosive attack on tubulars, particularly in the presence of H₂S-rich (sour) fluid. Hydrogen sulfide promotes sulfide stress cracking (SSC) and hydrogen embrittlement in steel, which increases the potential risk of sudden tubular collapse. "At temperatures above 150°C and pressures above 5,000 psi, H₂S dramatically lowers the threshold for sulfide stress cracking in common CT alloys." Consequently, CT strings must meet sour-service grades, include sufficient wall thickness, and undergo high-pressure tests to verify resistance (Table 1).

Table 1—Proper string design for HP/HT and high-H₂S wells, with segment selection and wall thickness based on the predicted well trajectory.

Wall Thickness (in)	Cu (WT%)	Ni (WT%)	Cr (WT%)	Mo (WT%)
.175	0.280	0.150	0.570	0.130
.175	0.280	0.160	0.590	0.140
.175	0.280	0.160	0.590	0.140
.175	0.280	0.160	0.590	0.140
.175	0.280	0.160	0.590	0.140
.175	0.280	0.160	0.590	0.140
.175	0.280	0.160	0.590	0.140
.175	0.280	0.160	0.590	0.140
.175-.190	0.280	0.150	0.570	0.130
.190-.204	0.280	0.170	0.630	0.140
.204-.190-.204	0.280	0.170	0.630	0.140
.204-.190-.204	0.280	0.170	0.630	0.140
.204-.190-.204	0.280	0.170	0.630	0.140
.204-.190-.204	0.300	0.160	0.570	0.140
.204-.224	0.300	0.160	0.570	0.140
.224-.204-.224	0.300	0.160	0.570	0.140
.224-.204-.224	0.300	0.160	0.570	0.140
.224-.204-.224	0.300	0.160	0.570	0.140
.224-.204-.224	0.300	0.160	0.570	0.140
.224-.204-.224	0.300	0.160	0.570	0.140

Solids Invasion and Near-Wellbore Skin

Drilling and completion operations introduce high-solids-load fluid, particularly those with weighted agents, into the formation. These fluids invade permeable streaks and deposit filter cake in less permeable or tight intervals. In HP/HT carbonate formations, these filter cakes commonly consist of calcium carbonate and bentonite. Temperature greater than 120°C triggers calcination and transforms the composition into a hardened, rigid barrier. This thermally altered filter cake forms a NWB skin that significantly impairs fluid communication with the formation.

Without prompt intervention, such as chemical breakers or acid wash treatments, this impermeable layer can persist beyond drilling and reduce the effectiveness of subsequent stimulation treatments. The restricted access to low-permeability zones prevents stimulation fluid from penetrating and enhancing formation conductivity, particularly in HP/HT environments where damage removal is already a challenge. Therefore, immediate post-drilling cleanup strategies are critical to restore NWB permeability and ensure optimal stimulation outcomes.

HP/HT Acid-Rock Interaction and Wormhole Propagation

A high bottomhole temperature (BHT) dramatically accelerates the HCl acid-CaCO₃ reaction rate. At 130°C, a 15% HCl acid solution consumes up to 90% of its volume dissolving at the rock face (facial dissolution), which limits wormhole penetration to only 1 ft. Low CT pump rates worsen the issue when required injection rates exceed CT mechanical limits. Under such conditions, a stimulation plan using 15% conventional HCl acid should shift to bullhead stimulation to achieve higher injection rates and a deeper wormhole formation.

Corrosion increases dramatically because of the high temperature and presence of HCl acid. The exothermic reaction between HCl acid and CaCO₃ (limestone) adds further heat, which intensifies the effect. At such conditions, the acid system should be equipped with a proper concentration of corrosion inhibitors and intensifiers to protect tubular integrity.

Fig. 4 shows the wormhole types formed during stimulation and the bypass of skin damage. The wormhole phase and penetration in the figure result from equal volumes of stimulation fluid. The wormhole structure directly depends on the acid volume required to achieve breakthrough in the core sample.

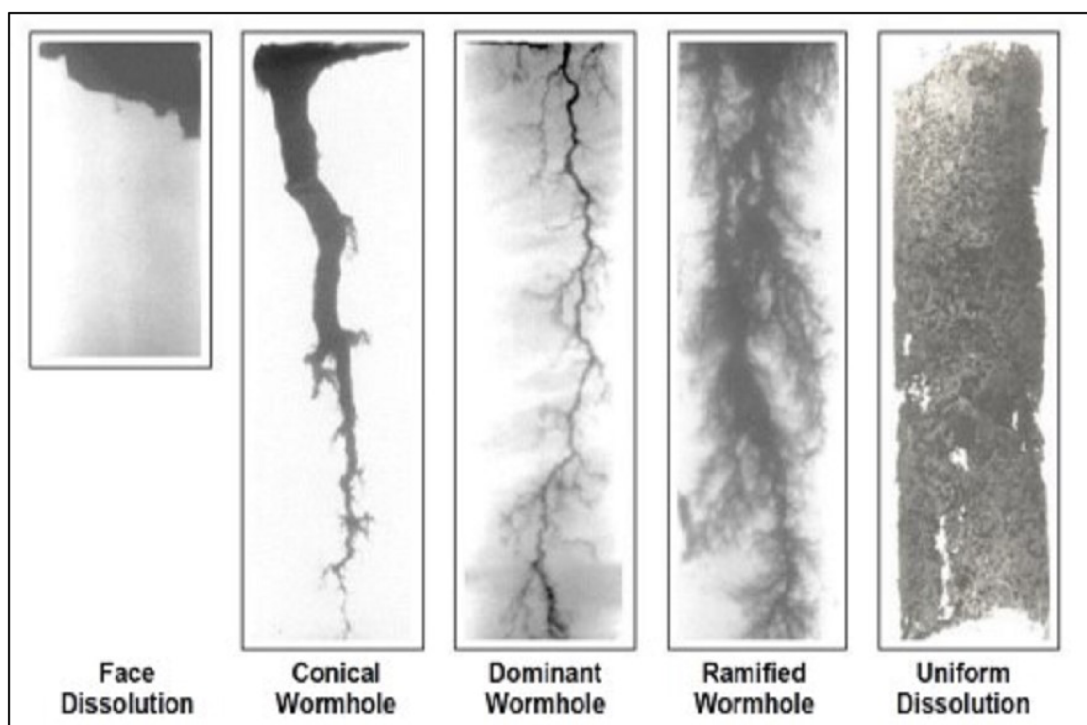


Figure 4—Dissolution structures observed in linear core tests (Fredd and Miller 2000).

The worst-case scenario in stimulation involves face dissolution. The best outcome involves uniform dissolution achieved through high injection rates and the use of retarded acid systems.

Fig. 5 illustrates the relationship among injection rates, temperature, and the wormhole phase. A higher temperature accelerates the acid reaction. Low acid injection rates cause face dissolution of the rock.

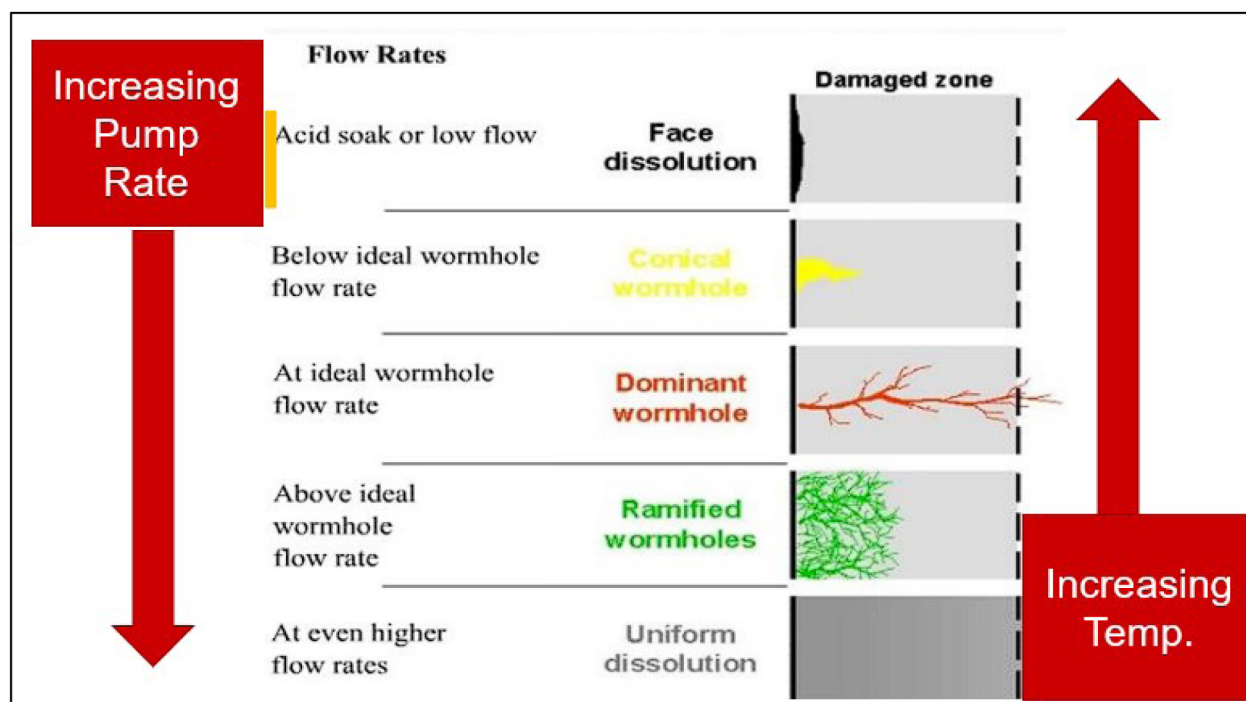


Figure 5—Relationship between temperature and acid injection rates in terms of wormhole penetration/phase.

On the contrary, high acid injection rates create better and more homogeneous dissolution along with temperature reduction. CT often cannot support high rates because of mechanical limits, friction pressure, and pump horsepower. Under high downhole temperatures, retarded acid systems or the bullhead mode provide better alternatives. If bullhead stimulation is selected to stimulate high-temperature formations, an injection test should confirm the maximum achievable rates before the introduction of the acid to the formation.

High Anisotropy Challenges

Heterogeneous carbonate reservoirs with high permeability contrast (anisotropy) present a major challenge for uniform acid distribution in the entire perforated interval. Without proper diversion, stimulation fluid flows into high-permeability zones and leaves the low-permeability or damaged intervals untreated.

Diversion Techniques to Overcome Anisotropy

The industry applies the following acid placement techniques to address this challenge.

Maximum Pressure Differential Injection Rate (MAPDIR). Increases the injection rate to force acid into high-skin or low-permeability layers through the increase of pressure drawdown in all zones.

Mechanical Diversion Through CT. Use CT to apply acid in stages or isolate zones with mechanical tools such as packers or jetting tools. Note: The dual injection mode can redirect fluid through a backpressure increase in the wellbore, forcing flow into low-pressure zones.

Chemical Diversion

Applies to systems that temporarily block high-permeability pathways:

- Linear and crosslinked gels
- Foam-based acids
- Bridging particulate systems

- Relative permeability modifiers (RPMs)
- Ball sealers (perf sealers)

Note: Some systems are not compatible with CT operations because of CT restrictions.

Straddle Packers. These mechanically isolate specific reservoir sections and enable zone-by-zone treatment. Once each layer is treated, operators reposition the straddle to the next interval.

Importance of Diversion

Effective diversion maximizes reservoir contact and ensures that stimulation benefits the entire reservoir thickness. According to Darcy's Law, the production rate increases in direct proportion to the effective reservoir height that contributes to the flow.

Example. A 10-m-thick reservoir includes:

- Upper 5 m with high permeability and high skin damage
- Lower 5 m with low permeability and high skin damage

Without diversion, acid enters mainly into the upper zone. The lower zone remains unstimulated, which results in only 50% of the reservoir that contributes to production.

With the application of diversion—mechanical, chemical, or both—the higher permeability zone is temporarily blocked. This redirects acid into the lower-permeability interval, which reduces skin and enables it to contribute to flow. The outcome includes increased well productivity and more uniform reservoir drainage, as described by Darcy's equation (Eq. 1).

$q = \frac{kh(\bar{p} - p_{wf})}{141.2B\mu \left[\ln(0.472 r_e / r_w) + s \right]}$	Eq. 1
$q = \frac{kh(\bar{P}^2 - P_w^2)}{1424qZT\bar{\mu} \left[\ln\left(\frac{0.472r_e}{r_w}\right) + S \right]}$	

Stimulation Treatment Design

The introduction outlined the challenges associated with this HP/HT reservoir. The next section presents specific data from the study well and defines a stimulation design that enhances and sustains productivity. This workflow includes diagnostics, treatment design, and post-operation simulation.

Reservoir Data

Table 2 through Table 4 and Fig. 6 summarize key reservoir characteristics, to include anticipated conditions, thickness, petrophysical properties, pressure, permeability, and flow behavior. These data highlight zone-by-zone variations and identify factors, such as high water saturation, skin damage, and asphaltene deposition, that impact production performance.

Table 2—Reservoir anticipated data.

Anticipated Reservoir Data	
Reservoir Type	Oil
Reservoir Pressure (psi)	8,500 psi
Reservoir Temperature (°F)	± 275
H ₂ S Content (ppm)	± 2,000
CO ₂ Content (mol%)	3%
Oil Gravity (API)	35 – 40
Anticipated GOR (avg. scf/bbl)	± 1,000 – 2,000
Shut in Wellhead Pressure (psi)	± 4,000
Flowing Wellhead Pressure (psi)	± 1,250
Flowing Wellhead Temperature (°F)	± 200

Table 3—Reservoir thickness and petrophysical properties indicate variations in porosity, with Zone 4 identified as a high water saturation zone.

Well	Zones	Top (m)	Bottom (m)	Gross (m)	Net (m)	Net to Gross	Av Shale Volume (v/v)	Av Porosity (v/v)	Av SW
SPE Well	1	4014.6	4037.39	22.79	14.40	0.63	0.01	0.16	0.22
	2	4037.39	4080.02	42.63	31.90	0.75	0.00	0.18	0.20
	3	4080.02	4125.44	45.62	3.80	0.08	0.00	0.11	0.35
	4	4125.44	4136.81	11.37	1.80	0.16	0.02	0.13	0.44

Table 4—Reservoir pressure and permeability data reveal significant variations between zones, both in pressure and permeability values.

No.	Zone	Top (m)	Bottom (m)	Permeability (md)	Pressure (psi)
1	1.1	4014.5	4021	2 – 10	6,000 – 7,500
2	1.2	4025.5	4033	50 – 100	6,000 – 7,500
3	1.3 / 2	4037.5	4072.5	5 – 10	7,200 – 7,600
4	3	4080	4083.5	0.5 – 1	7,000 – 7,500

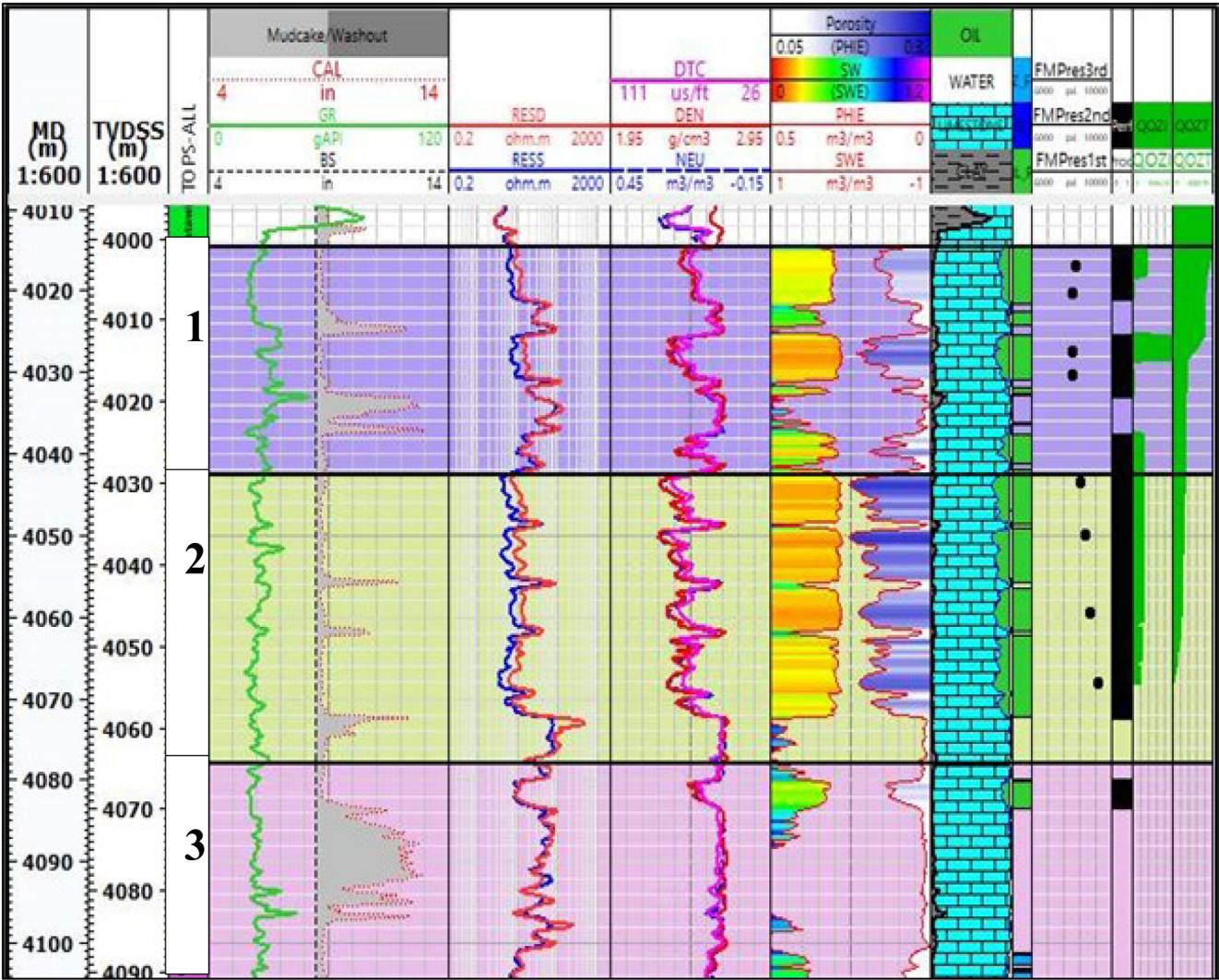


Figure 6—Well logs from the study show inconsistent flow behavior in a carbonate formation. Lower zones fail to produce oil despite high permeability. This discrepancy is attributed to severe skin damage and significant asphaltene precipitation in these zones.

Well Data

Tables 5 and 6 and Figs. 7 through 9 provide detailed well data, including history, structural profile, perforation specifications, production performance, and cement integrity. This information supports the diagnosis of flow restrictions, guides diversion strategy selection, and ensures acid placement during stimulation.

Table 5—The well history summarizes various slickline interventions conducted to collect samples and confirm the hold-up depth (HUD) at 4065 m. Multiple perforation intervals showed asphaltene blockage. The well was treated with a solvent system, which enabled deeper wireline access, followed by an additional intervention to address anticipated asphaltene deposition.

Well Intervention History		
Date	Operation	Summary
23/04/2018	Rigless wireline perforating	Perforation operations that were subsequently suspended because of loss of guns in the well and a loss of the primary wellbore barrier.
01/05/2020		
18/12/2018	Rigless pumping operations	Annular bullhead pumping operation with tracer dye to determine hole in completion. Result = leak at packer.
13/01/2020	Workover operations	Remove old/damaged completion and recomplete with a standard B9/Yamama completion + DHPG.
22/03/2020		
30/03/2021	Rigless intervention	Slickline: 2-in. bailer to 4065 m + 2.6-in. GC to same depth. Coiled tubing: CT solvent wash + 164 bbl of Yamama solvent squeeze into the perforated interval. Treatment was left to soak for 24 hr and then flowed back. Wireline dummy run to 4161 m, correlated to 3959 m. FGS and a two-rate + shut-in PLT.
09/20/2020	Rigless intervention	During CT RIH, 10 bbl of DAD was pumped with 4 hr soaking time and managed to pass. The well was opened, and CT fluid was displaced to 35 bbl of DAD at 3917 m during RIH. Matrix stimulation was performed from 4083 m to 4014 m in two passes. Maximum WHP was observed to be 1,260 psi.

Table 6—Well perforation details indicate that the well was perforated with high shot density and deep penetration, which provides valuable data for the selection of an appropriate diversion mechanism.

Perforations							
Carrier Size and Conveyance		Top Shot (m)	Bottom Shot (m)	Total Length (m)	Shots	Perforation Charges	
4.5-in. W/L 6 SPF 22.7 HMX 60 phasing 0.4-in. hole and 25-in. penetration		4014.50	4021.00	6.5	128	SPF	Phasing
		4025.50	4033.00	7.5	148		Charge Size
		4037.50	4072.50	35.0	689	6	60
		4080.00	4083.50	3.5	69		22.7 HMX
Total Net Interval:		52.5					
Interval Height:		69.0					

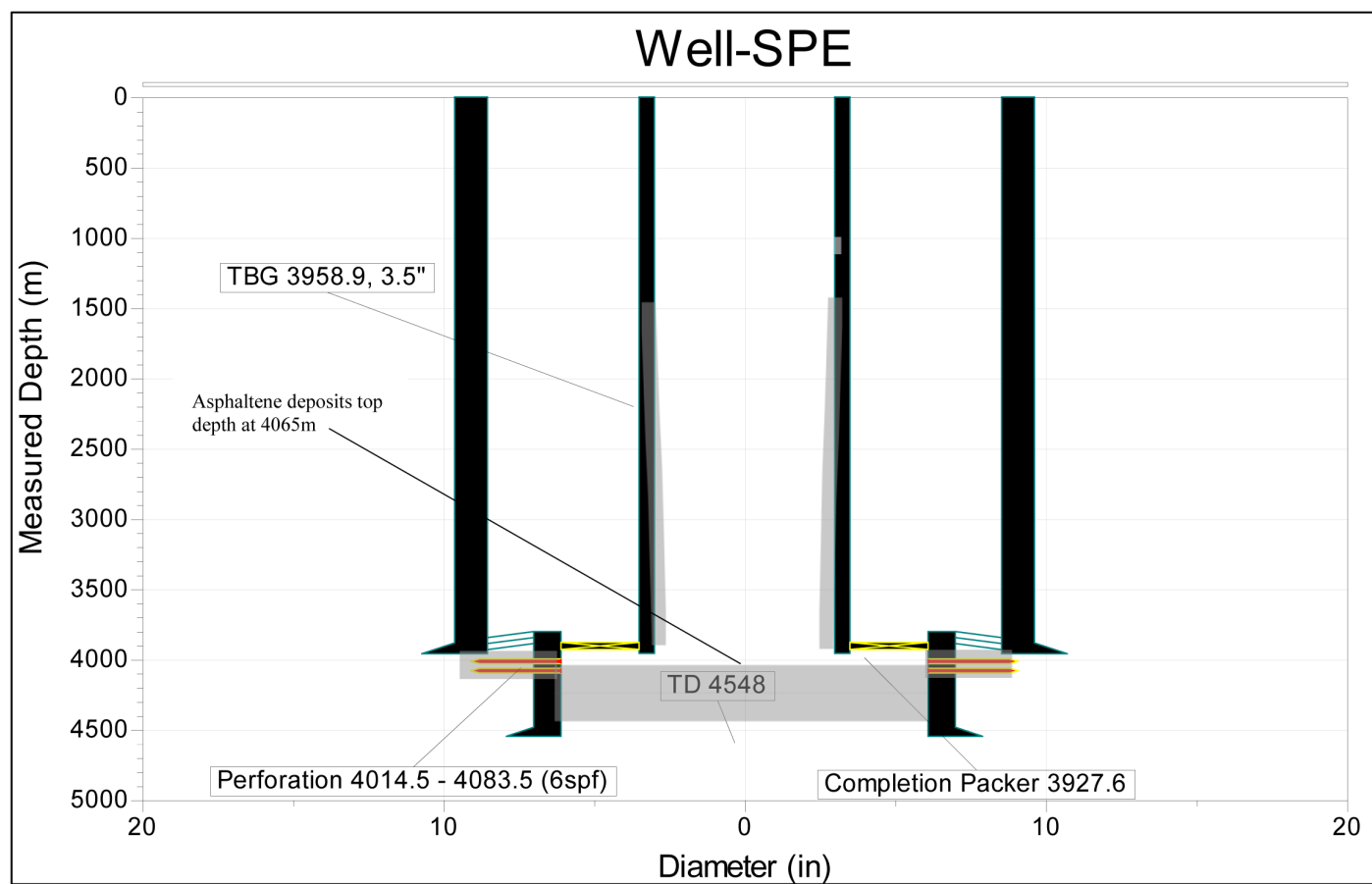


Figure 7—Structural profile of the study well that highlights zones affected by asphaltene deposition.

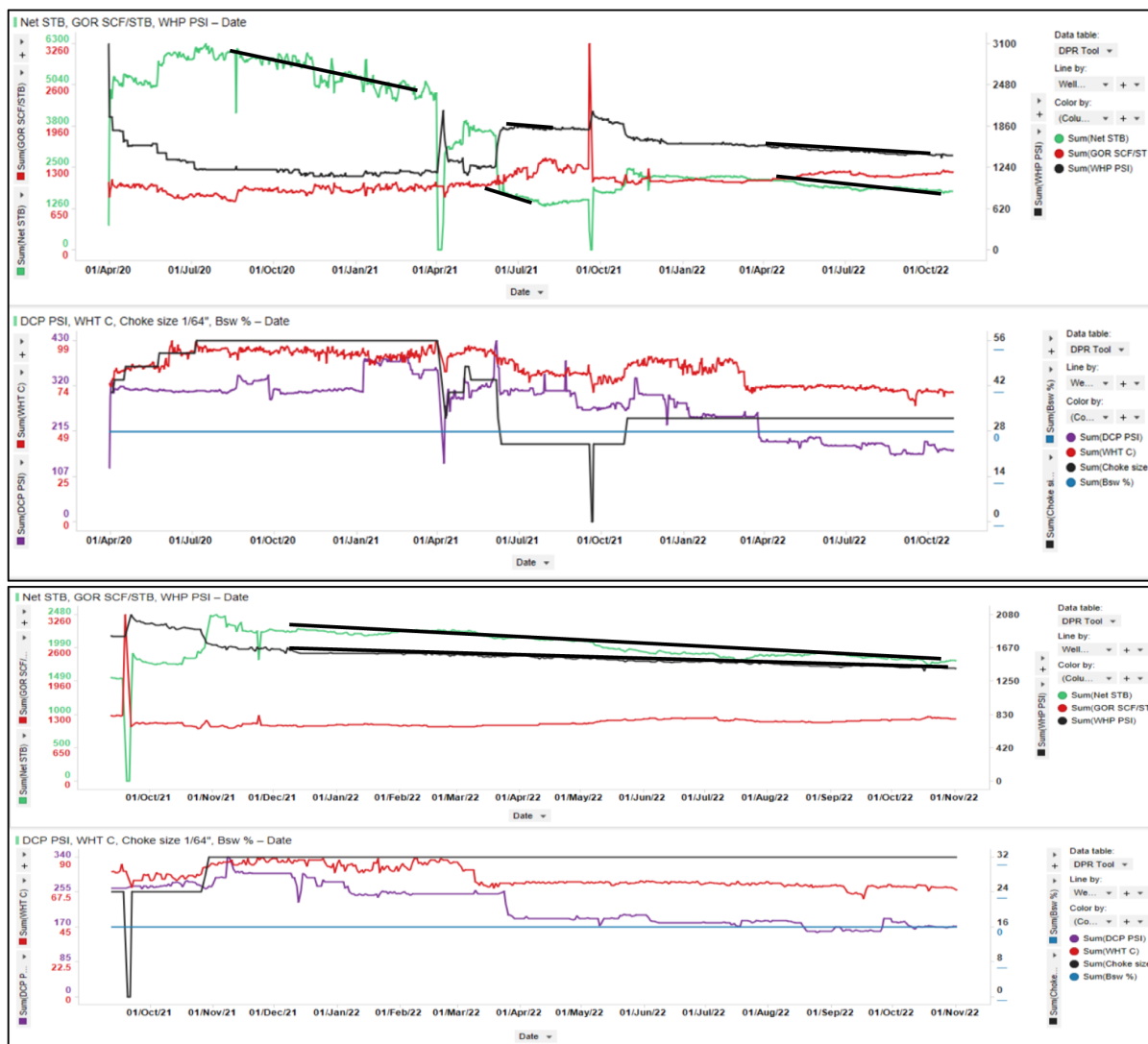


Figure 8—Production history illustrates a continuous decline in pressure and production over time, followed by noticeable improvements after each stimulation operation.

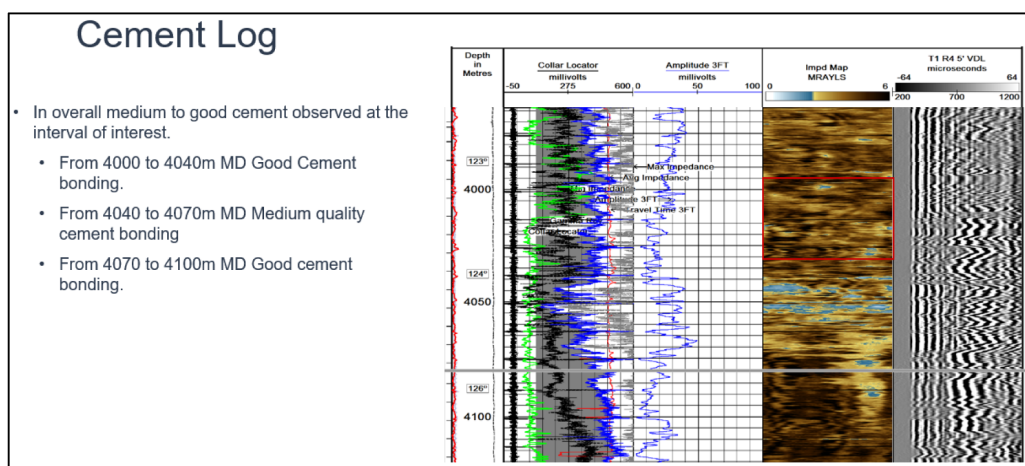


Figure 9—Cement bond log indicates good to medium cement integrity. During stimulation, proper evaluation of the cement behind the casing ensures that the acid reaches the formation and prevents unintended flow through cement-casing voids into unwanted zones.

Well Conditions Summary

A perforated well in an HP/HT reservoir is prone to asphaltene precipitation, which results in severe formation damage. Additionally, the combined effect of inorganic and organic skin further reduces production. Therefore, the well requires a comprehensive cleanup that removes both organic and inorganic deposits using an appropriate treatment system. Significant variability in permeability and porosity in the reservoir demands an optimized diversion strategy. The treatment design begins with the selection of an effective asphaltene removal technique using CT, including proper nozzle configuration and solvent system.

Asphaltene Removal

Traditional methods for asphaltene and paraffin removal include thermal treatments, mechanical removal, dispersants, and solvents.

Thermal Treatments

Hot oiling has been the most common technique for heat transmission downhole to remove paraffin deposits. However, while effective for paraffin, hot oiling can cause significant formation damage in the presence of asphaltenes ([Barker 1989](#)).

Mechanical Removal

Scrapers, cutters, and CT-deployed jetting tools provide the physical removal of deposits ([Akbar and Saleh 1989](#)). Mechanical removal is limited by its inability to address formation damage beyond perforation tunnels. Since the early 1990s, CT tools combined with solvents have improved removal efficiency ([Efthim et al. 1990](#)). Recent advancements claim improved performance against even the most severe asphaltene deposits ([Al-Ghazi and Lawson 2007](#)).

Dispersants

Water-soluble dispersants manage paraffin deposits through the dispersion of particles to enable circulation out of the wellbore ([Brogden and Dill 1969](#)). These chemicals do not dissolve paraffin or asphaltenes but assist in their physical removal. Performance improves with small additions of solvents such as heavy aromatic naphtha and with heating the make-up water temperature to dissolve paraffin and release co-deposited asphaltenes.

Solvents

Solvent stimulation of low-gravity oil reservoirs began in the late 1960s. Toluene and xylene served as the most commonly used solvents for asphaltene cleanup ([Jeffries-Harris and Coppel 1969](#)). Environmental concerns prompted the investigation of alternative solvents by the mid-1980s ([Houchin and Hudson 1986](#)). Laboratory-guided solvent selection emerged during the late 1980s ([King and Holman 1988](#)) and evolved into a systematic solvent design based on chemical theory in the 1990s ([Samuelson 1992](#); [Leontaritis et al. 1994](#); [Minssieux 2001](#)). Researchers evaluated cosolvents and refined blends, such as diesel and deasphalted oil, to enhance solvency ([de Boer et al. 1995](#); [Jamaluddin et al. 1996](#)).

Recent developments include environmentally friendly terpene-based solvents introduced in 2003 ([Curtis 2003](#)), which, while safer, do not match the solvency power of traditional aromatic solvents. Emulsified solvent systems (ESS) introduced in 2006 combine enhanced health, safety, and environmental (HSE) profiles with effective asphaltene solvation and improved water-wetting of the formation ([Lightford et al. 2006](#)).

Post-Cleanup Considerations

Improper pressure and temperature control after effective asphaltene removal during acid stimulation or wellbore access can cause reprecipitation. Since reservoir fluid composition is fixed, control focuses on the

minimization of thermodynamic conditions favorable for asphaltene flocculation and the modification of rock or pipe surfaces.

Surface properties, such as oily coatings, roughness, and electrical charge, influence deposit accumulation. Asphaltene inhibitors can coat surfaces to delay deposition, though this topic is beyond the current scope. Altering wettability toward water-wet conditions significantly reduces asphaltene adsorption (Rietjens and Steenbergen 2001). The stability of the water film depends on ionic strength and pH (Buckley et al. 1997). Water competes with asphaltenes for active adsorption sites and reduces overall deposition. Laboratory studies show that complete asphaltene removal is not feasible; however, partial removal combined with surface modification can effectively mitigate redeposition (Buckley et al. 1997). The success of any treatment hinges on the asphaltene removal process to enable the replacement of the asphaltene layer with water or inhibitors.

ESS vs. Conventional Solvents

Lightford et al. (2006) introduced a novel ESS to address the persistent production challenges caused by asphaltene deposition. Developed as a long-term treatment for wells producing paraffin- or asphaltene-rich crude oils, ESS combines superior asphaltene solvency with enhanced HSE characteristics.

Successfully applied in 2005 for the removal of paraffin and asphaltene deposits in southern Italy, the system is formulated as an emulsion of a water-surfactant solution blended with a high flash-point solvent and a cosolvent. This dual-action approach facilitates both dispersion and dissolution of asphaltene deposits more effectively than conventional solvents.

Key advantages of the ESS include:

- Flashpoint greater than 145°F (63°C), which enhance safety during transport and deployment
- Absence of benzene, ethylbenzene, toluene, and xylene (BETX) compounds, which reduces environmental and health risks
- High asphaltene dissolving power, augmented by the surfactant dispersion effects
- Formation remains water-wet, which delays asphaltene redeposition
- High water content reduces total chemical transport requirements

The application of an acid (water phase)/solvent emulsion system designed to treat both asphaltene deposits and inorganic materials entrapped within asphaltene chains is discussed. Asphaltenes, characterized by complex polyaromatic structures with heteroatoms and metals, vary significantly in molecular composition within crude oils. Consequently, an emulsified solvent recipe effective for one type of asphaltene might not perform equally well for another (Fig. 10). This variability underscores the critical importance of tailoring emulsified solvent formulations for each specific asphaltene deposit to ensure optimal removal efficiency.

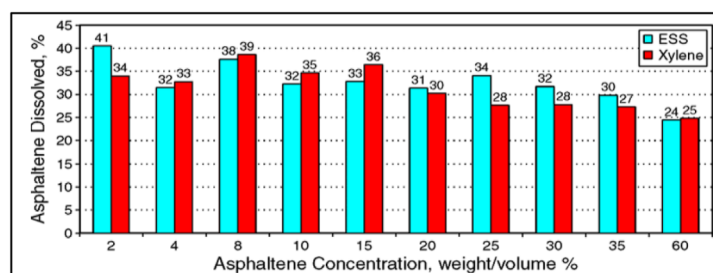


Figure 10—Comparison of ESS and xylene for maximum asphaltene dissolution using California asphaltenes.

Asphaltene Solubility Laboratory Test Procedure

Fig. 11 presents a sample treated with various solvents, which includes the ESS. A typical laboratory test follows the steps outlined below:

1. Measure the asphaltene sample and place it into a glass vial.
2. Add a precise aliquot of the ESS to the vial.
3. Seal the vial and shake it gently for approximately 15 seconds to promote dissolution and dispersion.
4. Slightly loosen the vial cap to help prevent pressure buildup during heating.
5. For comparison tests using xylene, apply the same procedure.
6. After the test duration, remove the vial from the water bath.
7. Filter the contents with a Buchner funnel and a weighed Whatman No. 42 ashless filter paper.
8. Wet the filter paper with isopropyl alcohol to enhance filtration efficiency.
9. Rinse the vial with isopropyl alcohol to transfer any undissolved asphaltene completely to the filter.
10. Rinse the undissolved material on the filter again with 5 to 10 mL of isopropyl alcohol.
11. Dry the filter paper and retain undissolved asphaltene in an oven.
12. Measure the weight difference of the filter paper before and after filtration to determine the mass of undissolved asphaltene.
13. Calculate the percentage solubility of the asphaltene using Eq. 2.

$$\% \text{ Solubility} = 100 - [(\text{undissolved asphaltene or original asphaltene}) \times 100] \quad \text{Eq. 2}$$



Figure 11—Sample collected from the well clearly showing asphaltene solidification.

Fluid used in the test are as follows:

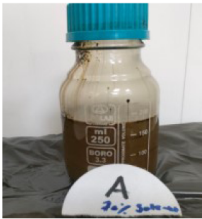
- ESS: 70% BETX-free solvents; 30% of HCl acid complete system (10%)
- ESS: 30% BETX-free solvents; 70% of HCl acid complete system (10%)
- Conventional solvents: 70% xylene and 30% musol

After following this procedure to assess the sample's solubility, Table 7 to Fig. 9 present the results.

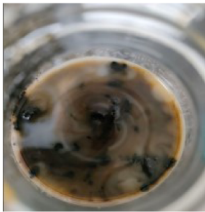
Table 7—ESS: 70% BETX-free solvents and 30% of HCl acid complete system (10% acid concentration).



ESS



ESS with 2 g of sample



The reaction: directly dissolved



Initial weight



Final weight



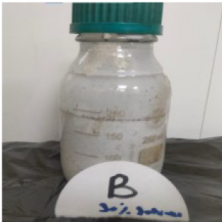
Residual amount

Sample	Fluids System	Wi (g)	Wf (g)	Solubility %	Remaining Percentage %
1	ESS 70% : 30%	2.0297	0.052	97.438	2.562

Table 8—ESS: 30% BETX-free solvents; 70% of HCl acid complete system (10%).



ESS



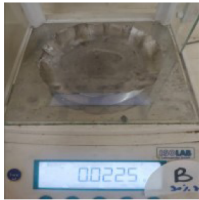
ESS and 2 g sample



The reaction: Not directly dissolved



Initial weight



Final weight



Reaming amount

Sample	Fluid System	Wi (g)	Wf (g)	Solubility%	Remaining percentage%
2	ESS 30% : 70%	2.0366	0.0225	98.8952	1.1048

Table 9—Conventional solvents: 70% xylene and 30% musol.

70% Xylene/ 30% Musol

Solution: 70% Xylene and 30% Musol
with 2 g of sample

The reaction: no reaction



Initial weight



Final weight



Reaming amount

Sample	Fluid System	Wi (g)	Wf (g)	Solubility%	Remaining percentage%
3	70% xylene and 30% musol	1.9988	0.0523	97.3834	2.6166

Laboratory Test Results and Conclusions

All fluid systems demonstrated strong solubility performance, as shown in [Table 10](#).

- The ESS formulation with 70% solvent and 30% acid fully dissolved the sample before placement in the heating bath.
- The ESS formulation with 30% solvent and 70% acid failed to dissolve the sample directly because of the high acid content in the system and dominant asphaltene composition.
- The conventional solvent system (70% xylene and 30% musol) immediately dissolved the sample. Slight residual particles precipitated, identified as calcium carbonate, and reacted visibly with HCl acid.

Table 10—Solubility performance.

Sample	Fluid System	Volume (ml)	Wi (g)	Wf (g)	Solubility (%)	Residual (%)
1	ESS (70% solvent and 30% HCl acid)	200	2.0297	0.052	97.4380	2.5620
2	ESS (30% Solvent and 70% Acid)	200	2.0366	0.0225	98.8952	1.1048
3	(Xylene 70% and Musol 30%)	200	1.9988	0.0523	97.3834	2.6166

A 2-gram sample quantity might not provide definitive formulation validation, yet these tests reveal how composition and solubility vary by fluid system and sample characteristics. For example, variations in calcium carbonate content could affect dissolution behaviors. Minor fluctuations in sample weight can also influence results.

Based on laboratory results, the ESS provides the most effective treatment solution. Its superior solubility, effective asphaltene inhibition, and ability to delay asphaltene redeposition and maintain a water-wet surface condition post-treatment make it the optimal choice for this design.

To further enhance asphaltene removal, it is recommended that the CT be equipped with a high-differential-pressure jetting nozzle. This setup will improve the mechanical force on asphaltene deposits along tubular surfaces and optimize stimulation effectiveness (Fig. 12).

		Flow Rate, bpm:		1.5	Fluid Density, lb/gal:		8.34	Pressure Drop, psi:		1783.69
Nozzle Set	Number of Nozzles	Diameter	Discharge Coefficient	Area Small	Velocity	Rate	Force			
		in		in ²	ft/min	bpm	lbf			
1 (Top)	1	0.125	0.80	0.010	24706	0.3750	28.018			
2	3	0.125	0.80	0.031	24706	1.1250	84.055			

Figure 12—Customized internal diameters optimize the tool's pressure drop. Downhole tools achieve maximum jetting efficiency through adjustments in nozzle diameter, nozzle count, and nozzle shape (e.g., rounded, hexagonal).

At elevated temperatures, the acid component of the ESS (aqueous phase) can be corrosive; however, with only a 10% acid concentration and the inclusion of an effective dosage of corrosion inhibitors and intensifiers in the formulation, the system maintains stability and demonstrates compatibility with the reservoir fluid.

Perforation Sealers Diversion Technique

Effective diversion ensures uniform wormhole development and compensates for reservoir anisotropy. Ball sealers are transported with the treatment fluid and are engineered to temporarily seal perforations that are actively taking fluid. This action diverts the treatment fluid to other perforations and promotes a more homogeneous stimulation. Theoretically, when ball sealers close all perforations, a "ballout" event triggers a significant pressure increase at the surface. When pumping stops and flow reverses, the ball sealers are released and fall into the rat hole.

Ball sealers apply to both acidizing and fracturing treatments and are available in various materials, densities, and sizes. The diversion strategy in the case study aligns with the following parameters:

- Reservoir temperature
- Anisotropy level
- Interval length
- Inflow characteristics
- Well trajectory

Based on these criteria, software calibration used flow percentages derived from production logging tool (PLT) data. Fig. 13 illustrates an interval with eight distinct zones, each with unique petrophysical properties and flow contributions. The fourth layer delivers the majority of production, while lower layers with higher permeability contribute minimally. Based on the following parameters, the most suitable diversion method is the use of ball sealers, as recommended by the diversion advisor tool (Fig. 14). However, it is important to note that there are various types of ball sealers available, each designed for specific applications and conditions.

Ball sealer selection depended on the following parameters:

- Perforation diameters
- Bottomhole temperature
- Differential pressure in the perforations

- Flow rate through the perforations
- Density contrast between the fluid and the ball

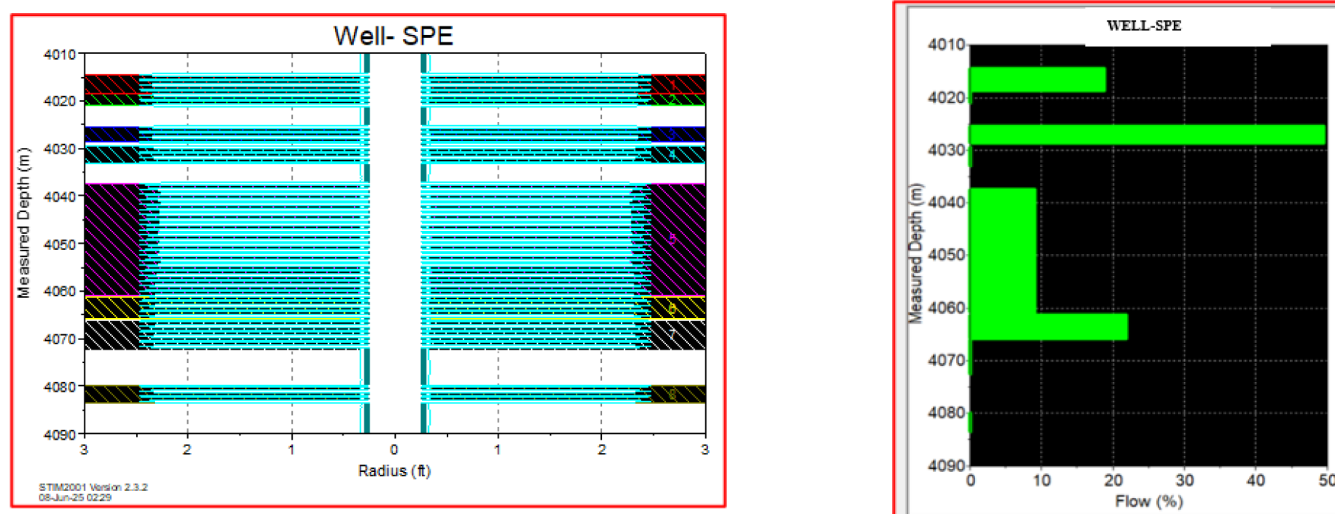


Figure 13—Well intervals along the hole with corresponding inflow percentages of each interval before stimulation.

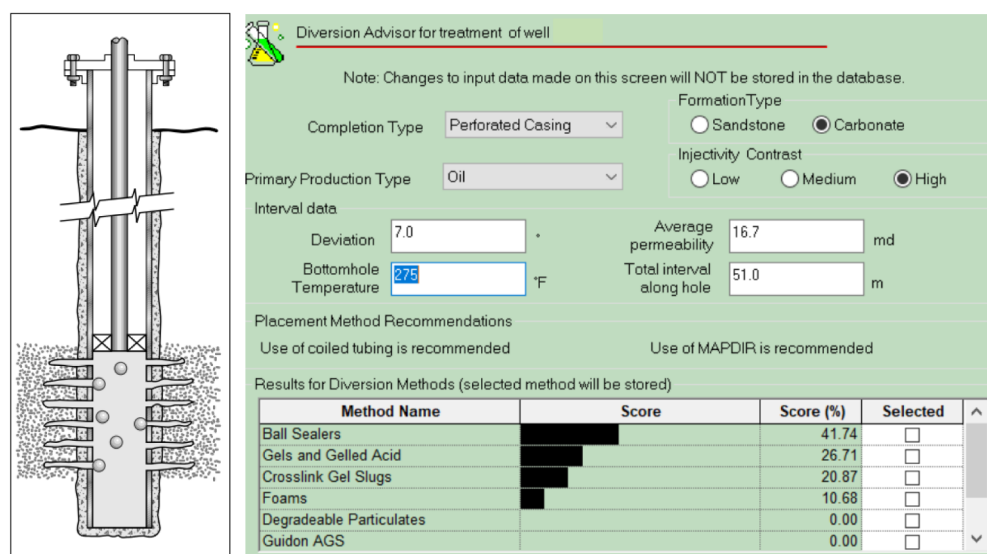


Figure 14—High-end diversion advisor software interface displaying performance scores for each diversion method based on formation characteristics.

General Guidelines

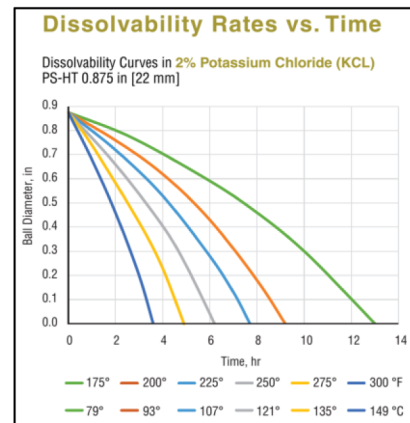
- Pipe inner diameter (ID) should be greater than three ball diameters for clustered drops; no restriction applies for singleball drops.
- Ball selection should match temperature and pressure differential ranges.
- Larger pipe diameters require higher pump rates to seat the ball.
- A greater density difference between the ball and the fluid 's specific gravities demands a higher pump rate to seat the ball.
- Higher fluid viscosity improves ball seating.

- Smaller balls seat more easily but must match the perforation size.
- Balls with a core diameter smaller than the perforations should not be used.
- Balls should exceed the perforation diameter by at least 1/4 in.
- Successful ball sealer use requires a strong cement operation and well-shaped perforations.
- Use excess nonbuoyant balls or sinkers, which typically range from 50 to 100% more than the number of perforations.
- Literature confirms 100% efficiency for buoyant balls at flow rates greater than 0.4 gal/min per perforation. The total number of buoyant balls used to divert a treatment should not exceed 110% of the perforation count.
- Soak the ball sealer in mineral oil, which helps prevent clumping, before gun or ball launcher loading.

The selection of rubber-coated phenolic/plastic (RCP) sinker balls followed these guidelines and criteria (Fig. 15). The carrier fluid has a specific gravity (SG) of 1.02, and the ball density is 1.2. This ball type is compatible with freshwater, brine, and acid; incompatible with nonpolar solvents such as diesel, benzene, xylene, kerosene, and toluene. It is suitable for fluid with pH values from 0 to 12 and is available in various sizes to match perforation diameters.

Selected Ball	Type	Ball Diameter	Maximum Perforation Diameter
		in	in
☐	RCA	0.875	0.520
☐	RCP	0.875	0.520
☒	RCP	0.875	0.520
☐	RCP	0.875	0.520
☐	RCN	0.750	0.520

(a)



(b)

Figure 15—(a) Selection of RCP balls based on ball diameter, maximum perforation diameter, and ball density; (b) work envelopes and solvency duration for the selected RCP balls.

Software recommendations and simulations eliminate the necessity for manual calculations and support accurate designs by covering the following parameters:

- Settling velocity of the balls
- Velocity of the balls
- Transport time of each ball
- Total pipe volume and fluid transport time
- Migration rate of each ball
- Verification of minimum stage volumes
- Ball injection schedule
- Required access of balls based on perforation count and ball density

Simulation

After software calibration with PLT data, the next step involves the creation of a pumping schedule and the simulation of the stimulation and diversion results with the selected ball type.

The design began with a conventional solvent system, followed by 100 gal/ft of 15% HCl acid with a friction reducer to minimize surface treating pressure. A spacer fluid was placed ahead of the ball sealer carrier and another spacer behind. The schedule included two diversion cycles (three total acid and solvent cycles), detailed in the pumping sequence (Table 11). The design incorporated several fluid types:

- Conventional Solvent Blend (70% xylene and 30% musol): Served as a spearhead fluid to dissolve organic material, such as residual asphaltene, in tubulars after CT stimulation with ESS or the asphaltene inside the formation.
- 15% HCl Acid: Functioned as the primary stimulation fluid; included a friction reducer to limit backpressure against surface equipment.
- Linear Gelled Fluid: A 35 cp gelled water used as both the spacer ahead of the ball sealers and as the carrier fluid and rear spacer.
- Treated Water: A 2% KCl water solution with surfactant and mutual solvent formulated to reduce the surface tension inside the formation and strip out the oil phase from the rock to increase the rock-acid interaction.

Table 11—CT treatment included perforation wash, well cleanup, and ESS-based stimulation to achieve 1.5 to 2 ft radial penetration in the formation.

Stage	Description	Fluid at Surface	CT Direction	Volume bbl	Rate bpm	CT Depth			CT Speed (m/min)	Pumping Time min	Choke Size, %	Total Time min
						From (m)	To (m)	Distance (m)				
1	Perforation Wash (Up)	ESS	POOH	20	1.4	4093	4005	-88	6	14	Open	14
2	Perforation Wash (Down)	ESS	RIH	20	1.4	4005	4093	88	6	14		29
3	Gel Sweep	Gell pill	Stop	5	1.4	4093	4093	0	0	4		32
4	Circulate 1.5 Bottoms up	SSC+H2S Scavenger blend	Stop	120	1.4	4093	4093	0	0	86		118
5	Adjusting choke manifold	ESS	POOH	40	1.4	4093	4005	-88	3	29	Closed	146
6	ESS Squeeze	ESS	RIH	40	1.4	4005	4093	88	-3	29		175
7	ESS Squeeze	SSC+H2S Scavenger blend	Stop	29	1.4	4093	4093	0	0	21		196

The designed treatment schedule shall include two ball diversion cycles. The design calls for shutoff of 75% of the perforation interval, which requires approximately 775 balls. To accommodate operational uncertainties and potential losses, an additional 10% excess is advised, which increases the total to approximately 870 balls. The design uses floating balls, which makes the excess necessary for an effective seal.

Because of limitations with the ball launcher equipment, two ball launchers that support 7/8-in. balls are available, with each unit designed for 250 balls. To compensate for this limitation, an additional ball launcher supports 5/8-in. balls. Consequently, the design incorporates both 7/8- and 5/8-in. ball sizes. The smaller balls offset the shortfall in the larger size. If sinker balls are used instead of floaters, the recommended excess drops to 50%.

The pump rate should exceed 5.5 barrels per minute (bbl/min) to provide the fluid velocity necessary for proper ball seating in the perforations. Based on the pumping schedule, simulation results confirm effective diversion and stimulation profiles, which successfully target the majority of the perforation interval.

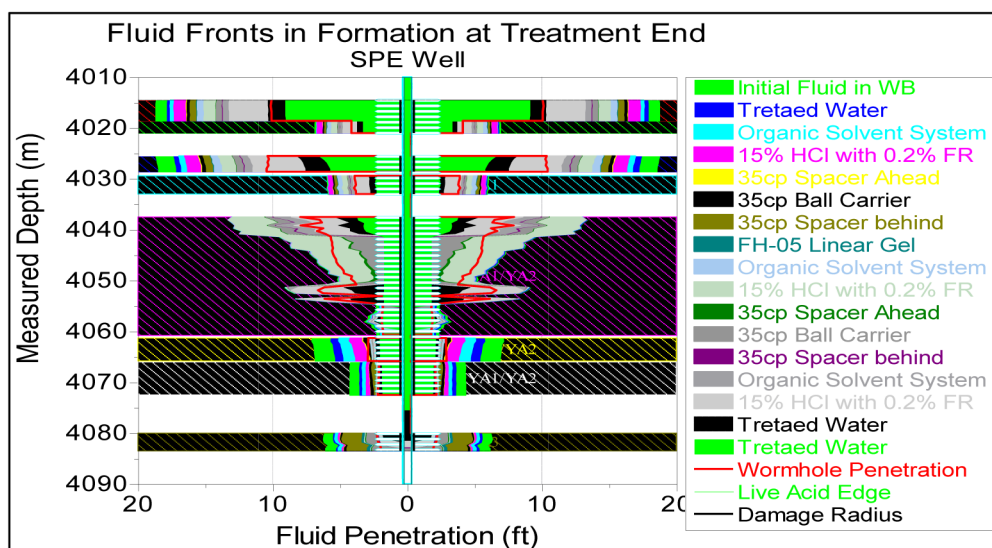


Figure 16—Fluid penetration at the conclusion of treatment and wormhole penetration within each zone, influenced by diversion efficiency and layer-specific characteristics. Radial wormhole penetration ranged from approximately 2 to 10 ft.

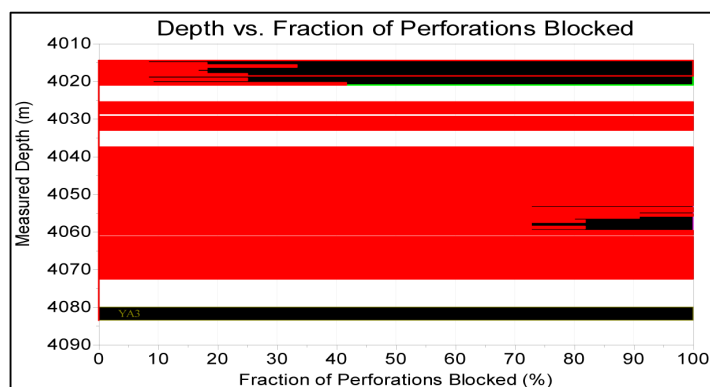
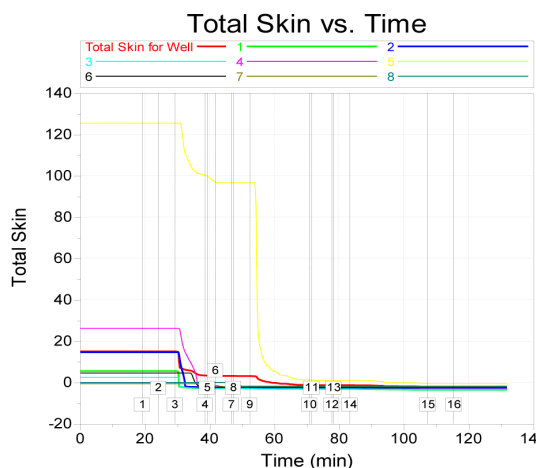
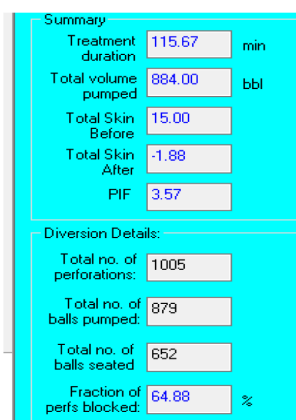


Figure 17—Perforation sealer blockage at each interval throughout the pumping period. Most perforations were successfully blocked by the end of the treatment, except for the last zone, which remained unblocked because of its lowest permeability and minimal inflow.



(a)



(b)

Figure 18—(a) Total skin damage calculated from interval inflow and permeability. Initial skin factor of 15 for this well decreased to -1.88 after stimulation, which raised the Productivity Index Factor to 3.57. (b) Percentage of perforations blocked by deployed ball sealers.

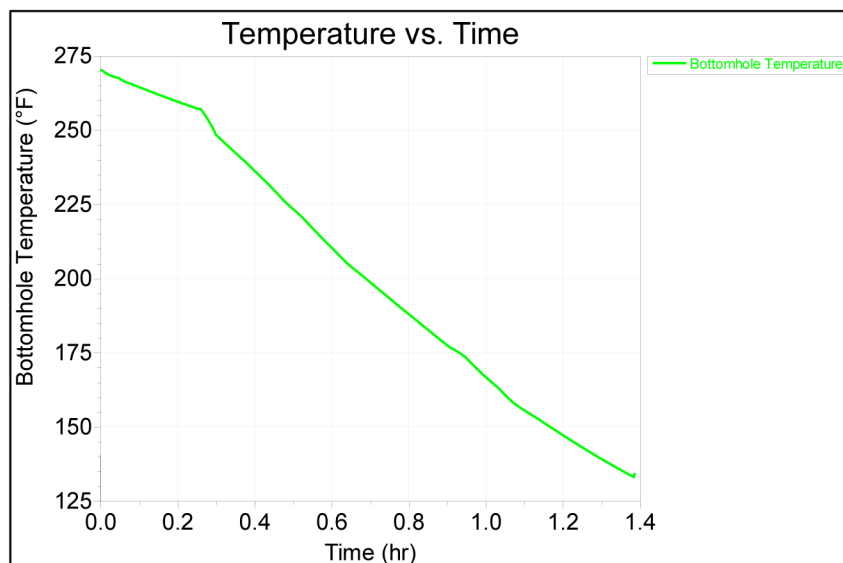


Figure 19—Temperature decrease caused by the pumping operation shows formation cooling from 275° to 135°F. This temperature drop defines the expected dissolution time for the ball sealers under these conditions.

A 12-hour shut-in period allows the ball sealers to dissolve adequately. At an average formation temperature near 200°F, the ball diameter is expected to decrease to approximately ± 0.3 in. during this interval.

Field Execution

The operation is divided into two phases: Well Accessibility and Well Stimulation.

Well Accessibility

This phase focused on the restoration of well access with the removal of asphaltene deposits and organic skin using CT and the ESS (Table 12). As a best practice, pickle the CT before pumping fluid into the formation by pumping 7.5% HCl acid displaced to the pit to ensure that no ferric residuals enter the well.

- Run-in-hole (RIH) CT to end of tubing (EOT) while pumping a blend of water, sulfide SSC inhibitor, and H₂S scavenger at a minimum rate.
- At EOT, circulate out 10 bbl of ESS followed by 5 bbl of 35 cp gelled fluid. Recover the pill at surface by bottoms up with brine at maximum rate. This process clears organic and inorganic buildup from the tubing, while the gelled fluid is considered the bed for any solids/undissolved materials and helps prevent fallback into the wellbore.
- RIH CT to HUD at 4050 m, then reach total depth (TD) at 4150 m, 66.5 m below bottom perforation. Observe tag points at 4075 and 4122 m. Perform pull tests and apply simulated force analysis weight. Raise the nozzle jet pressure with an increase in the pump rate of the SSC and H₂S scavenger blend to pass through asphaltene obstructions. At TD, pump 15 bbl of gel and circulate the well clean with the SSC and H₂S scavenger blend.
- Perform casing pickling using 15 bbl of 15% HCl acid followed by 10 bbl of gel sweep. Circulate all fluids to the surface.
- Perform a perforation wash with ESS while reciprocating CT in the perforation interval at maximum allowable rate. To manage the HP/HT reservoir and perform a perforation wash, apply backpressure (wellhead pressure) of 100 to 400 psi with an adjustment to the choke size to force the stimulation system to penetrate the first inches of the formation and flow up to the surface.

- At the end of the perforation wash, perform a gel sweep followed by $1.5 \times$ bottoms up of the SSC and H₂S scavenger blend. Once the return is observed clean, shut in the well.
- With the well shut and CT full of ESS, reciprocate the CT in the perforation while pumping ESS at a maximum rate achievable within wellhead pressure and CT limitations.
- During the operation, observe severe wellhead pressure fluctuations caused by asphaltene deposition within tubulars and inside the formation.

Table 12—Implemented SUT and SDT on the injection test.

Injectivity Test		
Rate (bbl/min)	Fluid	Time (min)
1	FR treated water	0.5 – 1.0
2	FR treated water	0.5 – 1.0
4	FR treated water	0.5 – 1.0
6	FR treated water	0.5 – 1.0
8	FR treated water	0.5 – 1.0
10	FR treated water	0.5 – 1.0
8	SDT	0.5
6	SDT	0.5
4	SDT	0.5

This phase successfully restored well access and enabled a clear injection path for the future bullhead stimulation.

Well Stimulation

Bullhead stimulation used a conventional solvent spearhead, 15% HCl acid with a friction reducer as the stimulation fluid, and perforation sealers as the diversion method, carried by 35 cp gelled fluid.

Best practices include injection tests before bullhead stimulation to confirm the injectivity index, defined as the maximum achievable rate within acceptable applied wellhead pressure (WHP). Test results showed a maximum pressure of 3,000 psi at 10 bbl/min, which confirmed readiness to proceed with the stimulation (Fig. 20).

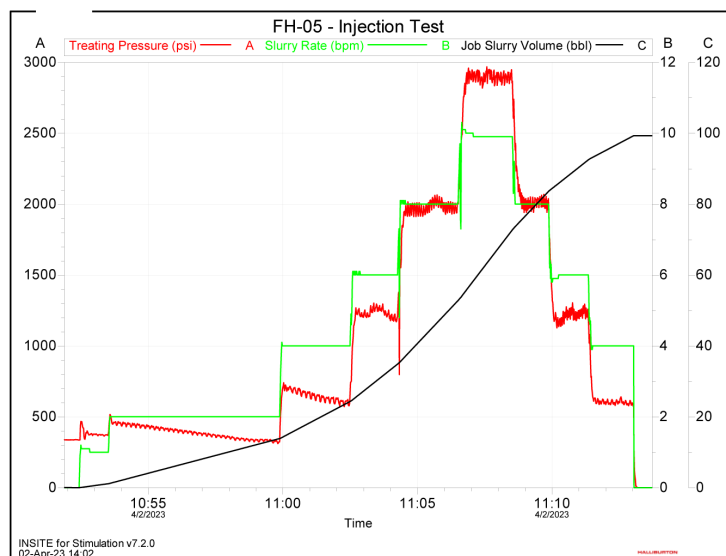


Figure 20—Corresponding pressure with the designed Table 11.

Perforation blocking requires knowledge of the effective perforation count, maximum achievable surface injection rate, fracture extension rate, and pressure (Fig. 21).

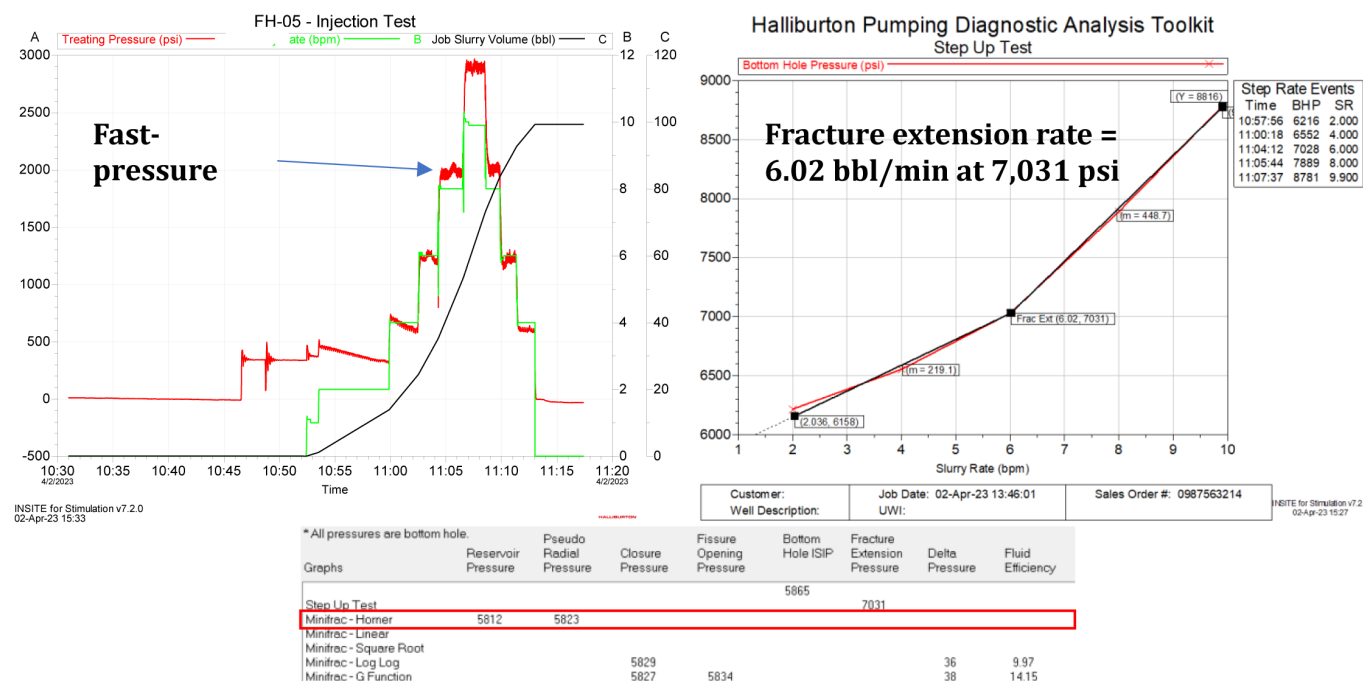


Figure 21—Fracture extension pressure and rate indicate the initiation point of fracture. A Horner plot allows the estimation of current reservoir pressure based on this test.

- **Step-Up Rate Test (SUT):** This test confirms the maximum injection rate achievable within surface equipment limits and identifies the fracture extension rate and pressure.
- **Step-Down Test (SDT):** This test calculates the entry friction, which includes both perforation friction and NWB friction (Fig. 22 and Fig. 23).

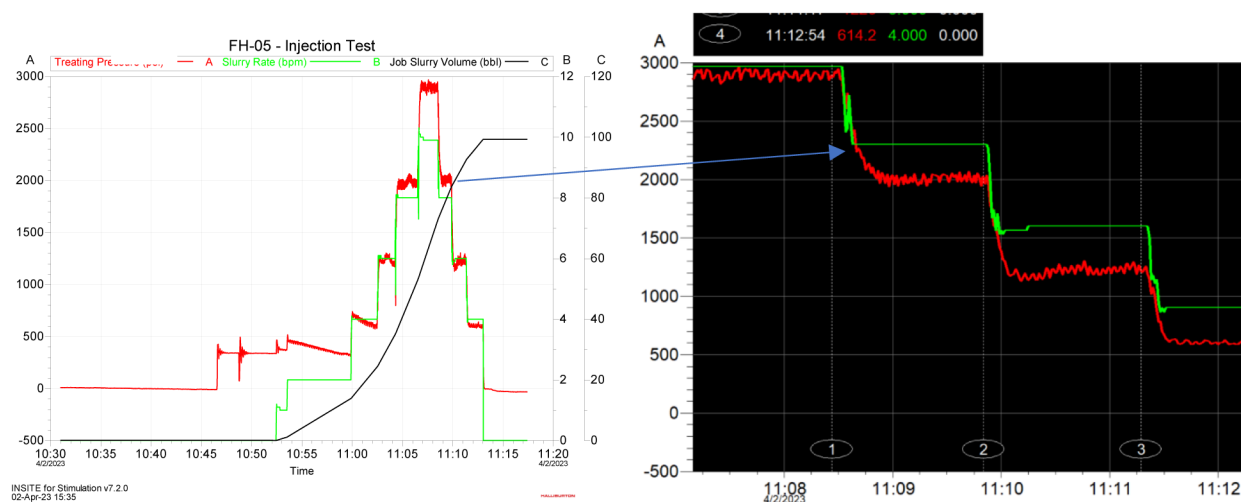


Figure 22—Instantaneous shut-in pressure (ISIP) observed during the SDT. Elevated ISIP indicates poor reservoir-wellbore connectivity caused by formation tortuosity and NWB damage.

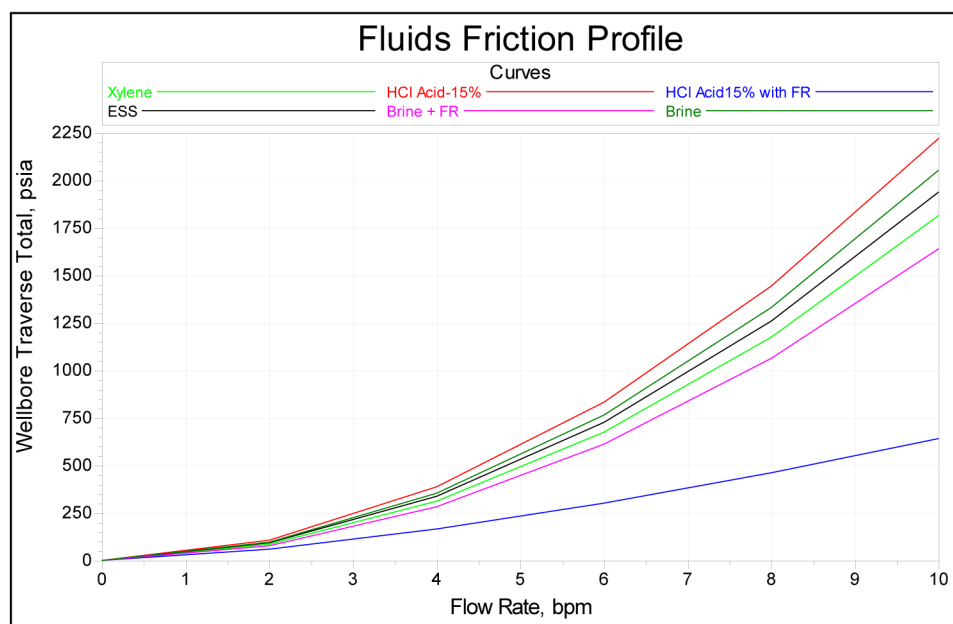


Figure 23—Effect of tubular friction at various rates and with different fluids, summarized for each fluid system used in the treatment as evaluated by high-end software.

Injection test data established operational limits at 8 to 10 bbl/min and 5,000-psi WHP.

- Initiated injection with treated water that contains 2% KCl and surfactant at 8 bbl/min.
- Pumped conventional solvent as a spearhead to dissolve the organic skin inside the formation.
- Pumped 15% HCl acid with a friction reducer to reduce excessive pressure from fluid-metal interaction.
- Pumped gelled fluid ahead of the balls, as a carrier fluid, and as a spacer behind.
- Repeated the full cycle.
- Overdisplaced fluid into the formation with treated water that contains 2% KCl and surfactant.
- Shut in the well for 24 hours to allow the ball sealers to degrade under downhole temperature.

Fig. 24 presents the operation chart.

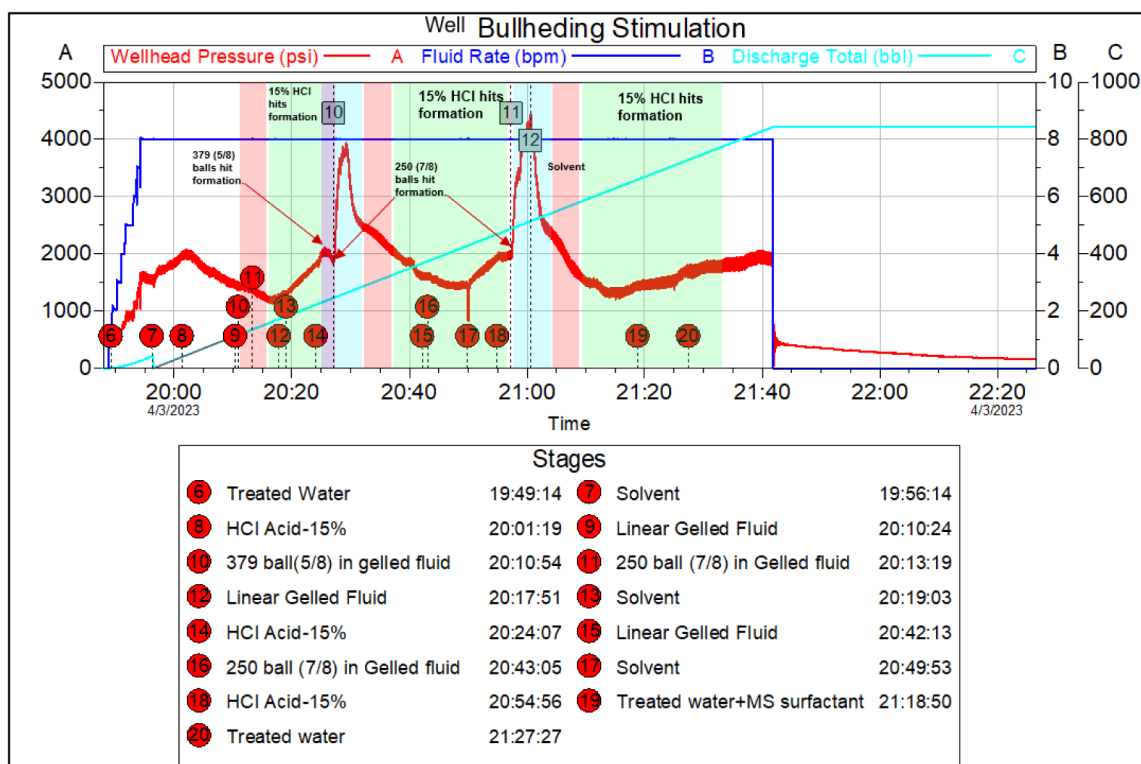


Figure 24—Summary of surface operation stages with corresponding downhole pressure. Notable pressure spikes indicated the seating of 7/8-in. balls.

Execution Summary

Fig. 24 confirms sustained and maximized oil production because it has targeted most of the reservoir thickness. Previous steps preserved water-wet conditions in the formation and delivered inhibitors with the ESS, which entered the reservoir and delayed asphaltene precipitation.

Several key observations appear in the chart:

- Pressure began to decrease at 20:02 when the solvent reached the formation. The drop resulted from reduced friction between the two fluid types and differences in their densities.
- Pressure shifts during acid contact with the formation resulted from wellbore density changes and frictional effects. These shifts caused an increase in WHP during acid fill, followed by a drop during pumping and displacement with the lower-density solvent. This behavior aligns with Eq. 3.

$$\text{WHP} = \text{PBHP} - \text{Phydro} - \text{Pfriction}$$

Eq. 3

- The 7/8-in. ball sealers achieved clear diversion success, while the 5/8-in. balls delivered weaker results. This outcome confirms the critical importance of ball size compatibility with perforation diameter (Fig. 25).
- Pressure remained below the 5,000-psi threshold, which confirmed the selected ball quantity.
- The final ISIP showed favorable fluctuations compared to the injection test, which indicated restored connectivity between the reservoir and wellbore because of effective damage removal.
- The linear gel reduced the ball settling velocity from 0.33 ft/sec to approximately 0.06 ft/sec. This improvement resulted from adjustments to gel SG and rheology using additives such as KCl and CMHPG.

- The solvent stage, placed ahead of the acid, enhanced the cleanup of newly created flow paths from organic deposits, which facilitated improved acid contact and reaction in previously unstimulated zones (Fig. 26 and Fig. 27).

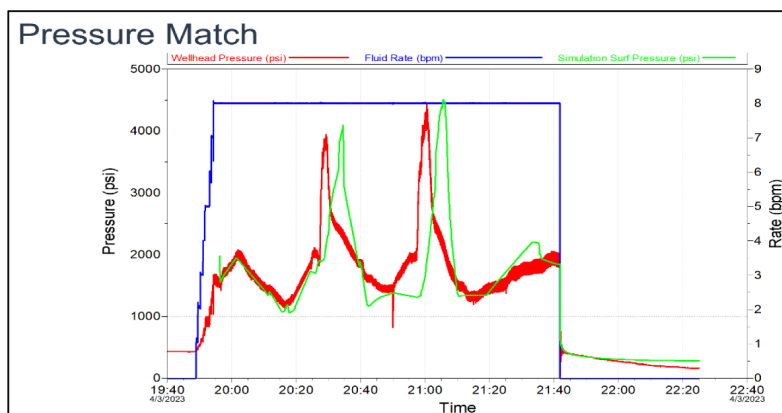


Figure 25—Pressure matching is essential to accurately estimate key reservoir parameters, such as skin factor, permeability, porosity, and inflow performance after stimulation.

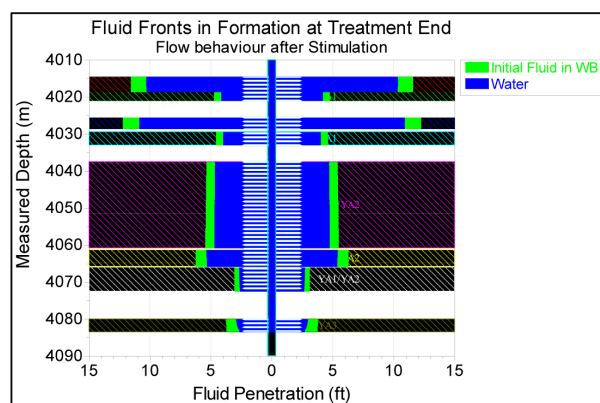


Figure 26—Post-stimulation fluid placement based on updated inflow percentages, which reflect changes in skin factor, permeability, and porosity.

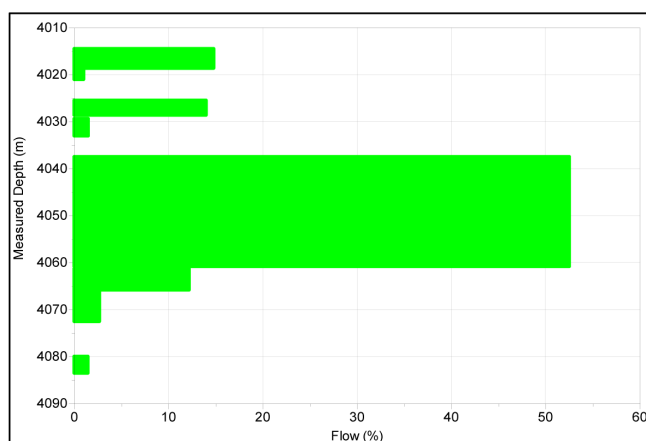


Figure 27—Flow contribution after stimulation based on pressure matching, which shows a significant increase in the longest interval compared to the initial well condition (PLT).

Results of this Study Well

During operation, a comparison between the stimulation ISIP and injection test ISIP revealed an evident indication of improved reservoir-wellbore connectivity. The water hammer effects confirmed the successful removal of skin damage.

Two pressure spikes indicated an effective diversion in the formation. The subsequent pressure drop confirmed successful stimulation and skin reduction.

After stimulation, the well returned to production. Oil output increased from 1,700 to 2,500 BOPD, which demonstrates a significant improvement.

Post-Stimulation Production Data

Fig. 28 and Fig. 29 show post-stimulation well production data.

Ball sealer diversion techniques saw wide use in multiple oilfields with complex reservoir parameters. The following sections present several operations executed under HP/HT conditions using ball sealers as the diversion method, with each case demonstrating effective changes in production profiles (Fig. 30 and Fig. 31).

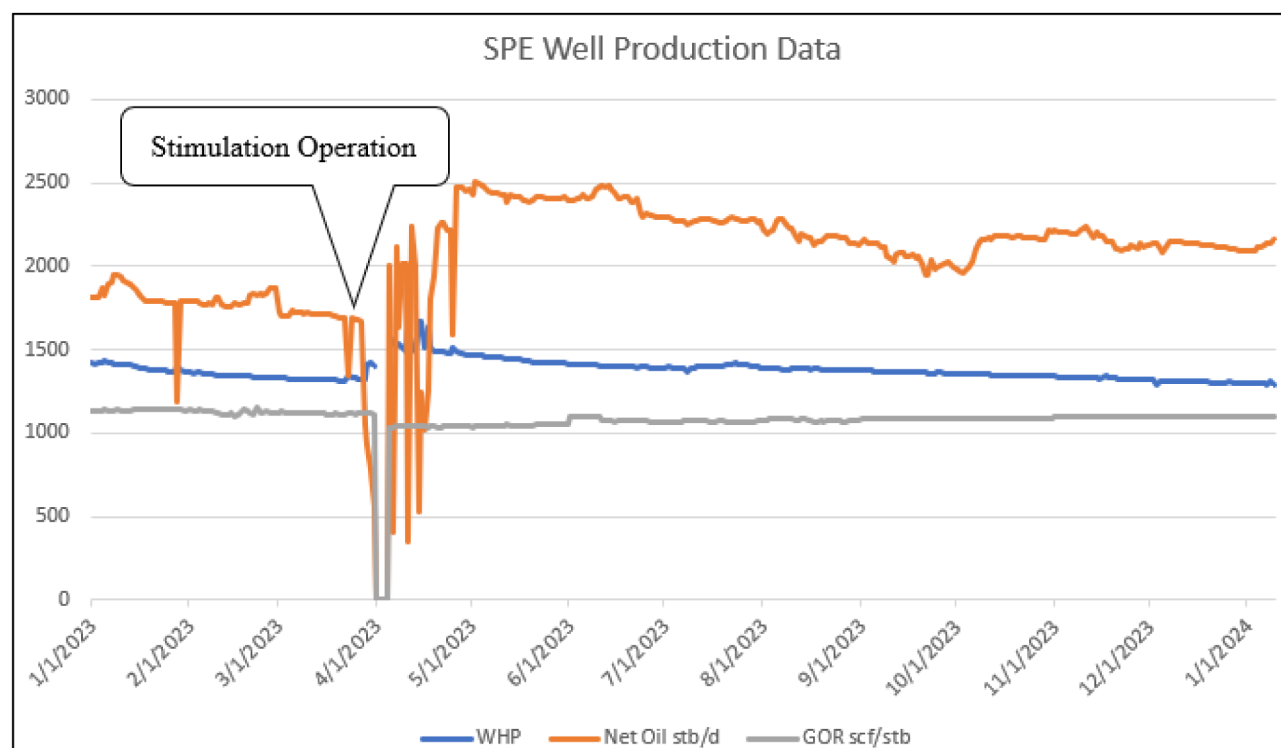


Figure 28—Production data indicates increased rates and stable production over multiple months.

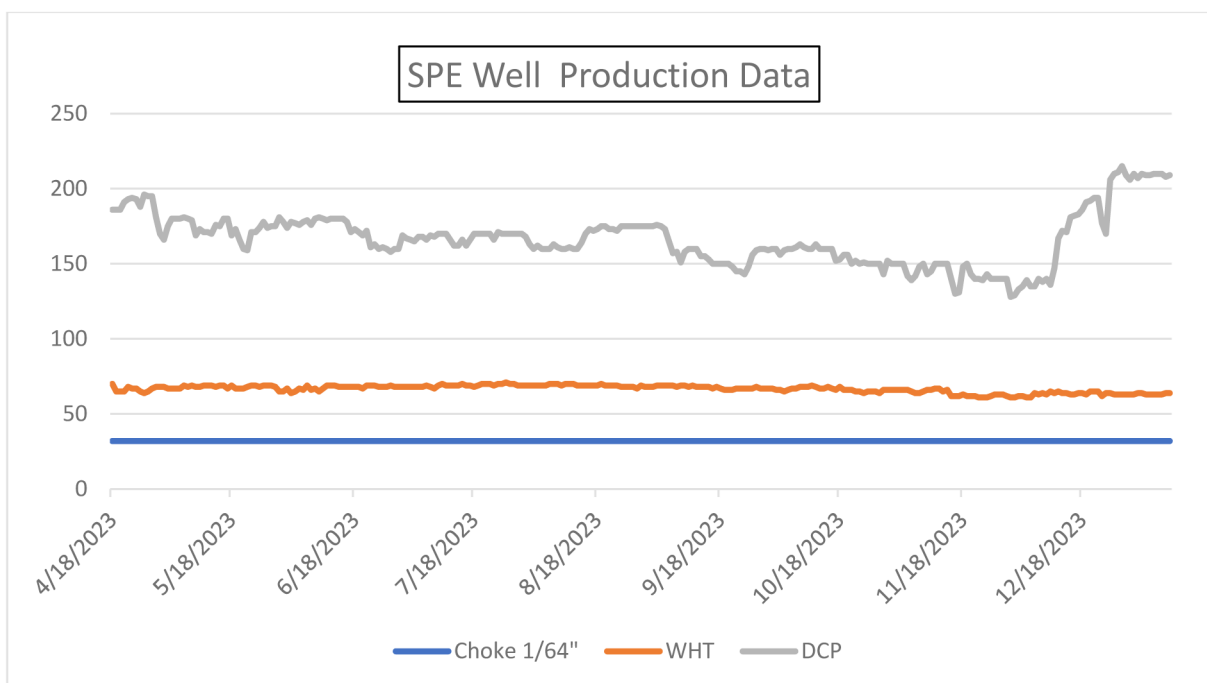


Figure 29—Production data recorded with choke size fixed at 32/64 in.

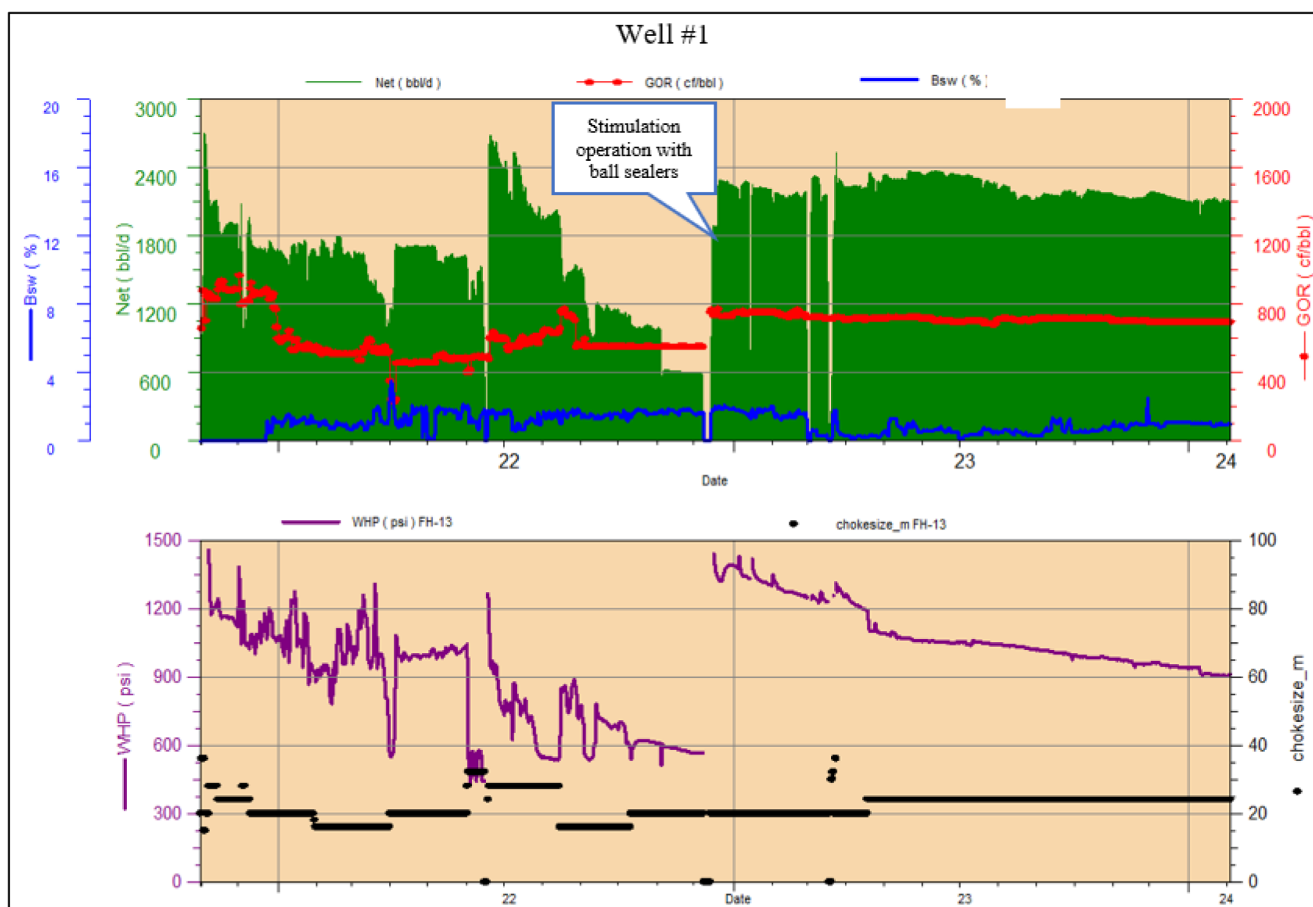


Figure 30—Well 1 production data at a fixed choke size of 32/64 in. indicates successful stimulation of a HP/HT zone, which resulted in increased production and wellhead pressure.

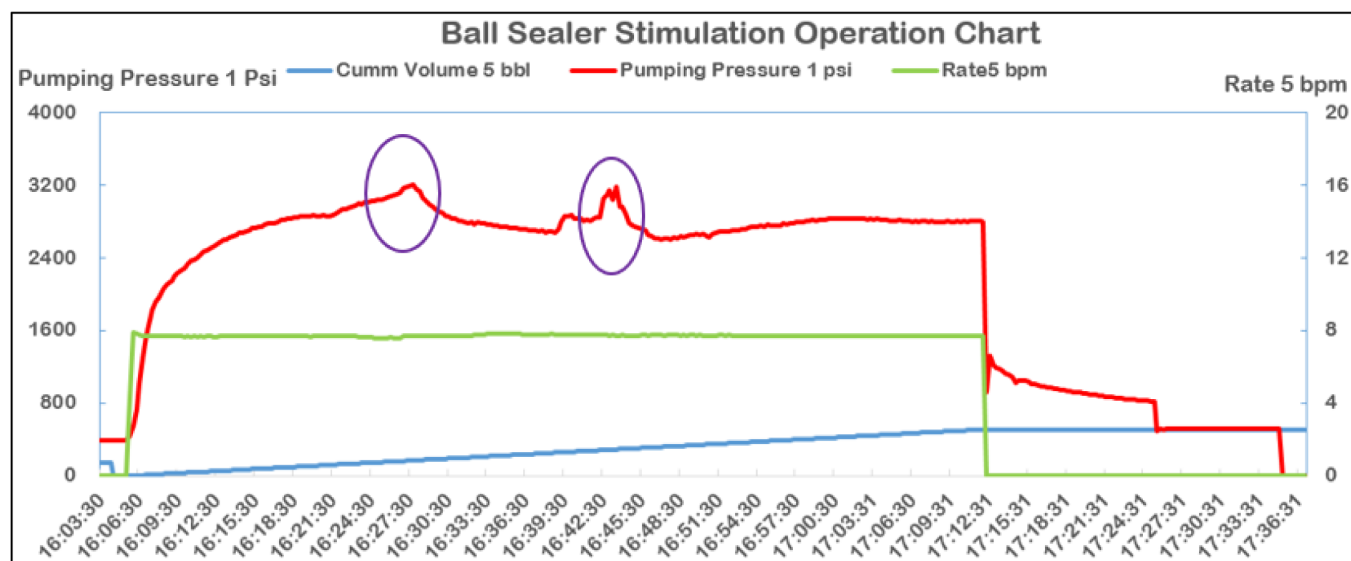


Figure 31—Operation chart for Well 1 indicates two diversion cycles based on pressure response.

Well 2 had HP/HT conditions and high anisotropy, which created diversion challenges; the well was newly drilled and had not undergone prior stimulation (Fig. 32, Fig. 33, and Table 11Table 13).

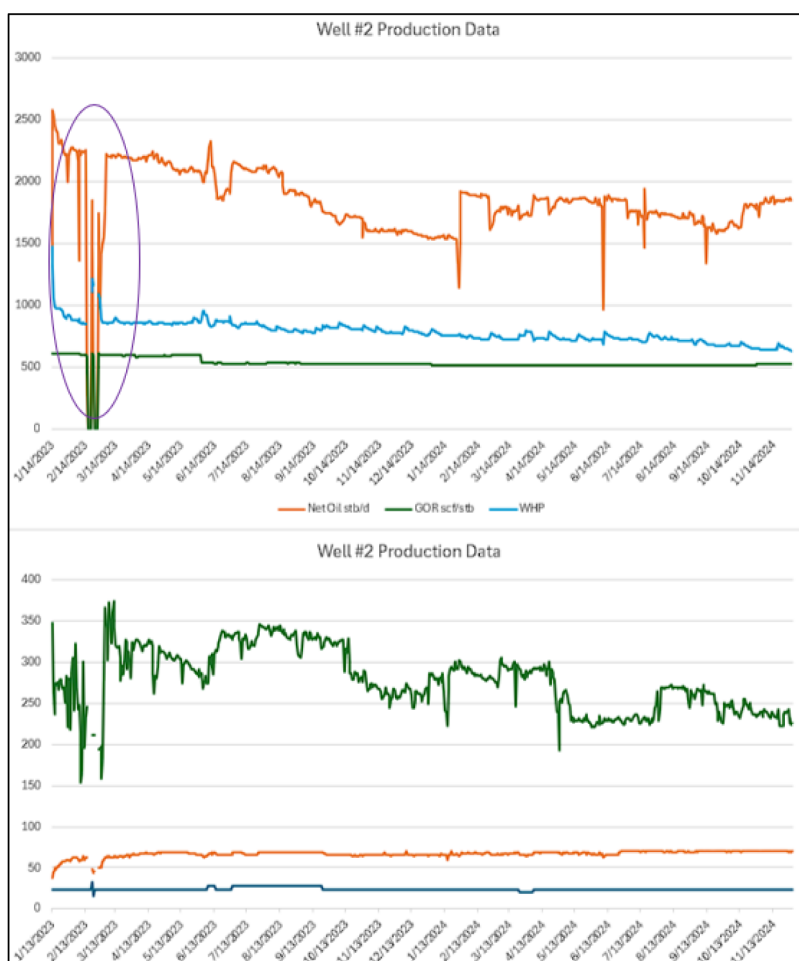


Figure 32—Production performance before and after the operation.

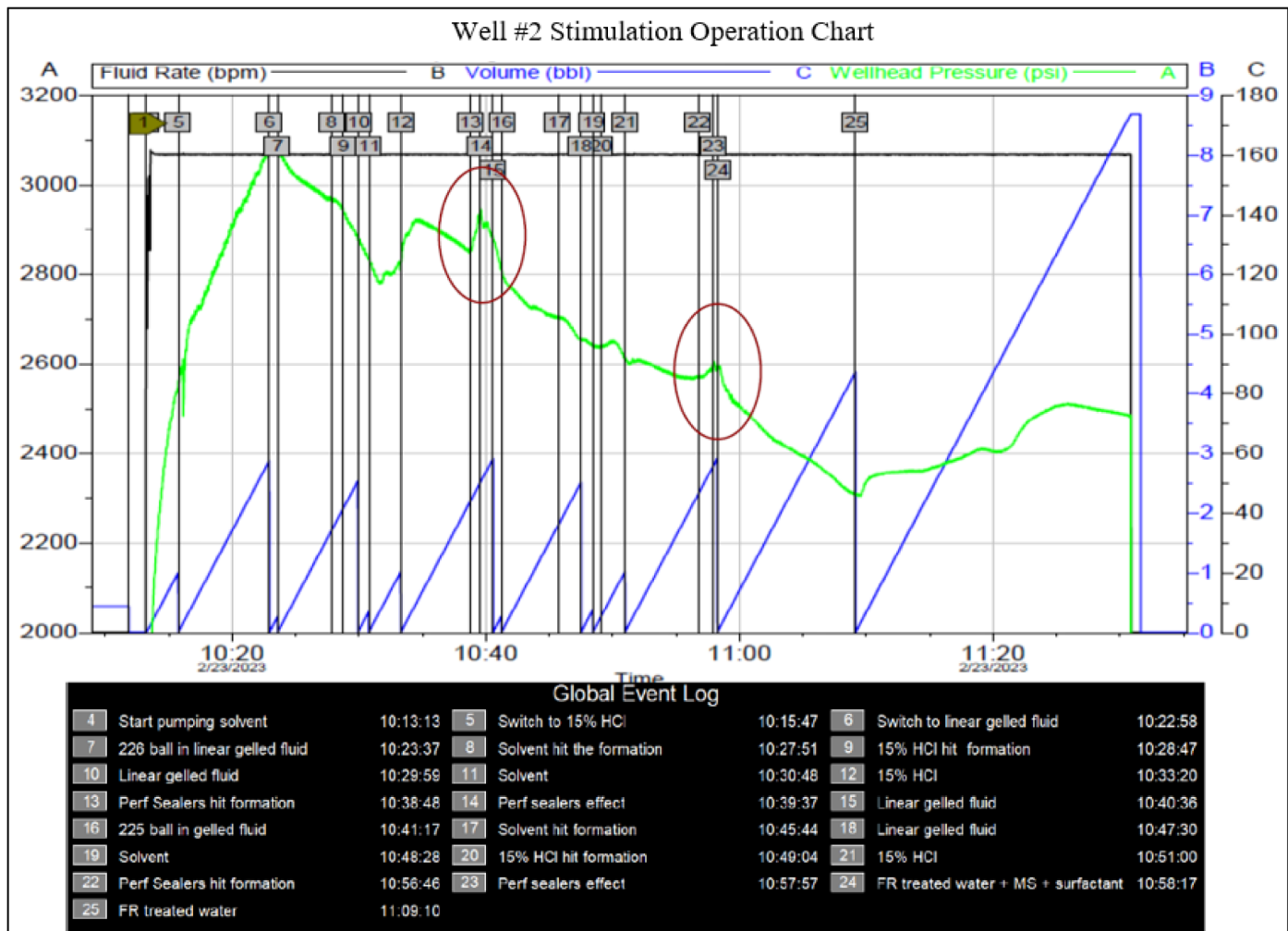


Figure 33—Operation chart illustrates acid effectiveness in damage removal and the sequence of diversion cycles.

Table 13—Summary of production data before and after stimulation.

Well	Production Before Stimulation (BOPD)	Production After Stimulation (BOPD)
SPE	1,700	2,500
1	600	2,200
2	1,500 – 2,500	2,250

Conclusion

A complete stimulation workflow should address oil market challenges with the introduction of methods that reduce the cost per barrel of produced oil and maximize well productivity in a sustainable manner.

Stimulation systems with dual functionality, such as asphaltene removal, delay of asphaltene precipitation, and rock stimulation, that operate at CT fluid rates represent a modern and effective approach.

High pump rates with fast-reaction acid systems are essential to achieve deep wormholes and to reduce the face dissolution of the rock. High rates can also promote diversion; however, without proper diversion, the entire operation risks failure. A combination of traditional and advanced diversion techniques ensures proper acid distribution in the formation.

All fluids, particulates, balls, or foams used as diverters shall undergo validation through tests to confirm that they do not cause formation damage.

The stimulation workflow should begin with accurate diagnostics of well damage, followed by a tailored design with proper tools and parameters. Precise execution ensures success and peak well performance, supported by design inputs based on reliable stimulation software.

Future recommendations include the use of continuous temperature and pressure measurements of the formation, adjustments to solvent formulations to match asphaltene variability, and the integration of real-time data analytics to improve treatment effectiveness and operational efficiency.

Nomenclature

ADP	asphaltene disappearance pressure
AOP	asphaltene onset pressure
BETX	benzine ethylbenzene toluene xylene
BH	bullheading
BHT	bottomhole temperatures
BOPD	barrel oil per day
CT	coiled tubing
DLA	diffusion-limited aggregation
ESS	emulsified solvent system
EOT	end of tubing
HP/HT	high-pressure/high-temperature
ISIP	instantaneous shut-in pressure
HS&E	health safety and environment
HUD	hold-up depth
MAPDIR	maximum pressure differential injection rate
NWB	near-wellbore
PBHP	bottomhole pumping pressure
PLT	production logging tool
RCP	rubber-coated phenolic/plastic
RIH	run-in-hole
RLA	reaction-limited aggregation
RPM	relative permeability modifiers
SSC	sulfide stress cracking
SDT	step-down test
SL	slickline
SUT	step-up rate test
TD	target depth
WHP	wellhead pressure

References

- Akbar, S.H. and Saleh, A.A. 1989. A Comprehensive Approach to Solve Asphaltene Deposition Problem in Some Deep Wells. Paper SPE 17965 presented at the SPE Middle East Oil and Technical Conference and Exhibition, Manama, Bahrain, 11-14 March. <https://doi.org/10.2118/17965-MS>.
- Al-Ghazi, A.S. and Lawson, J. 2007. Asphaltene Cleanout Using VibraBlaster Tool. Presented at the SPE Saudi Arabia Section Technical Symposium, Dhahran, Saudi Arabia, 7-8 May. <https://doi.org/10.2118/110972-MS>.
- Al-Safran, E.M., Aql, A., and Ghasemi, M. 2025. Investigating Asphaltene Deposition Behavior During Carbon Dioxide (CO₂) Injection Using the Asphaltene Integrated Model (AIM). Presented at the SPE Conference at Oman Petroleum & Energy Show, Muscat, Oman, 12-14 May. <https://doi.org/10.2118/224842-MS>.
- Barker, K.M. 1989. Formation Damage Related to Hot Oiling. *SPE Prod Eng.* 4 (04): 371-375. <https://doi.org/10.2118/16230-PA>.

- Brogdon, B.R. and Dill, F.E. 1969. A New Chemical Approach to Stimulation of Old Producing Wells. Presented at Fall Meeting of the Society of Petroleum Engineers of AIME, Denver, Colorado, USA, 28 September-1 October. <https://doi.org/10.2118/2671-MS>.
- de Boer, R.B., Leerlooyer, K., Eigner, M.R.P., et al 1995. Screening of Crude Oils for Asphalt Precipitation: Theory, Practice and the Selection of Inhibitors. *SPE Prod & Fac.* **10** (01): 55-61. <https://doi.org/10.2118/24987-PA>.
- Efthim, F.P., Garner, J.J., Bilden, D.M., et al 1990. Evaluation and Treatment of Organic and Inorganic Damage in an Unconsolidated Asphaltic Crude Reservoir. Presented at the SPE Formation Damage Control Symposium, Lafayette, Louisiana, USA, 22-23 February. <https://doi.org/10.2118/19412-MS>.
- Erbstoesser, S.R. 1980. Improved Ball Sealer Diversion. *J Pet Technol.* **32** (11): 1903-1910. <https://doi.org/10.2118/8401-PA>.
- Fredd, C.N. and Miller, M.J. 2000. Validation of Carbonate Matrix Stimulation Models. Presented at the SPE International Symposium on Formation Damage Control, Lafayette, Louisiana, USA, 23-24 February. <https://doi.org/10.2118/58713-MS>.
- Frost, K.A., Daussin, R.D., and van Domelen, M.S. 2008. New, Highly Effective Asphaltene Removal System With Favorable HSE Characteristics. Presented at the SPE International Symposium and Exhibition on Formation Damage Control, Lafayette, Louisiana, USA, 13-15 February. <https://doi.org/10.2118/Z112420-MS>.
- Ghaffoori, A., Anderson, R., Hoopanah, M., et al 2023. Successful Application of a Novel High-Pressure High-Temperature (HPHT) Quartz Crystal Microbalance (QCM) Setup to the Evaluation of Asphaltene Deposition Inhibitors. Presented at the Middle East Oil, Gas and Geosciences Show, Manama, Bahrain, 19-21 February. <https://doi.org/10.2118/213750-MS>.
- Lightford, S., Pitoni, E., Arnesi, F., et al 2006. Development and Field Use of Novel Solvent-Water Emulsion for the Removal of Asphaltene Deposits in Fracture Carbonate Formations. Presented at the SPE Annual Technical Conference and Exhibition, San Antonio, Texas, USA, 24-27 September. <https://doi.org/10.2118/101022-MS>.
- Nozaki, M., Zhu, D., and Hill, A.D.D. 2013. Experimental and Field Data Analyses of Ball-Sealer Diversion. *SPE Prod & Oper.* **28** (03): 286-295. <https://doi.org/10.2118/147632-PA>.
- Salgaonkar, L. and Danait, A. 2012. Environmentally Acceptable Emulsion System: An Effective Approach for Removal of Asphaltene Deposits. Presented at the SPE Saudi Arabia Section Technical Symposium and Exhibition, Al-Khobar, Saudi Arabia, 8-11 April. <https://doi.org/10.2118/160877-MS>.
- Yen, T.F. and Chilingarian, G.V. 1994. Introduction in *Asphaltenes and Asphalts*, ed. G.V. Chilingarian and T.F. Yen, Chap. 1, **16**. Amsterdam: Elsevier Science.
- Jeffries-Harris, M J., and Coppel, C P. (1969). Solvent Stimulation in Low Gravity Oil Reservoirs. *Society of Petroleum Engineers*. <https://doi.org/10.2118/2158-PA>.
- Houchin, L.R. and Hudson, L.M. 1986. The Prediction, Evaluation, and Treatment of Formation Damage Caused by Organic Deposition. Presented at the SPE Formation Damage Control Symposium, Lafayette, Louisiana, 26-27 February 1986. SPE-14818- MS. <http://dx.doi.org/10.2118/14818-MS>
- King, G. E. and Holman G. B., Jr., "Hydrocarbon Solvents: An Alternative to Acid for Removing Some Formation Damage," paper SPE 14136 presented at the 1986 International Meeting on Petroleum Engineering, Beijing, China, 17-20 Mar.
- Samuelson, M. L., "Alternatives to Aromatics for Solvency of Organic Deposits," paper SPE 23816 presented at the 1992 SPE International Symposium on Formation Damage Control, Lafayette, Louisiana, 26-27 Mar.
- Lightford, S., Pitoni, E., Arnesi, F., and Mauri, L., "Development and Field Use of Novel Solvent-Water Emulsion for the Removal of Asphaltene Deposits in Fracture Carbonate Formations," paper SPE 101022 presented at the 2006 SPE Annual Technical Conference and Exhibition, San Antonio, Texas, 24-27 Sep.
- Rietjens, M. and Steenbergen, P., "Clean Up of Asphaltene Damaged Wells for Sonatrach," Halliburton B.V. European Research Centre Report HTECX09001-L006, Leiderdorp, The Netherlands, 12 June, 2001.
- Buckley, J. S., Liu, Y., Xie, X., and Morrow, N. R., "Asphaltenes and Crude Oil Wetting—The Effect of Oil Composition," *SPEJ*, **2**, 107-119 (1997).